

Geometric digital twinning of full-scale metal shells of revolution – modelling protocol and buckling-relevant fabrication tolerance assessment

Lijithan Kathirkamanathan^a, Adam J. Sadowski^a & Marc Seidel^b

^a *Department of Civil and Environmental Engineering, Imperial College London, UK*

^b *Siemens Gamesa Renewable Energy, Germany*

Corresponding author: Adam J. Sadowski, Email: a.sadowski@imperial.ac.uk

Abstract

The structural response of fabricated full-scale metal shells of revolution such as chimneys, masts, monopiles, silos, tanks and wind turbine support towers (WTSTs) is known to be exceptionally sensitive to their actual constructed geometry which thus requires a high-fidelity representation for accurate modelling. Within aerospace, mechanical and civil engineering, ‘geometric imperfection’ surveys of both laboratory-scale manufactured specimens and full-scale fabricated shells have long been conducted to characterise imperfection forms for improved structural resistance assessments. However, the methods employed as part of point-cloud-to-model software protocols have varied greatly over the years and a consistent framework has always been lacking, hindering automation and compelling research teams to reinvent the wheel by redeveloping tools and software afresh.

This paper presents a standardised modularised protocol for the systematic processing of reality capture surveys of shells of revolution in point cloud format, bringing together historical and recent developments in data analysis to perform various processing steps including enhanced registration, can segmentation, systematic and random outlier removal, surface reconstruction, surface inpainting and projection to arbitrary meshes. The protocol facilitates the automated generation of arbitrary meshed representations for post-construction quality assessment, damage and lifetime extension evaluations, and advanced structural analysis with real geometries. Novel algorithms for automatically assessing EN 1993-1-6 (2025) buckling-relevant fabrication tolerances (dimple, out-of-roundness and unintended eccentricity) are presented. The protocol is illustrated on a real high-definition WTST laser scan and is readily generalisable to other metal shells of revolution. A 5.7 GB augmented sandbox WTST point cloud dataset and implementation scripts written in MATLAB are provided.

Keywords

Geometric digital twin, fabrication tolerance quality assessment, geometric imperfections, geospatial survey, structural health assessment, metal wind turbine support towers.

1. Introduction

The accurate structural response of full-scale thin-walled metal shells such as chimneys, masts, monopiles, silos, tanks and wind turbine support towers (WTSTs) is controlled to the greatest extent by their actual ‘imperfect’ constructed surface geometry which can be significantly different from the nominal or ‘perfect’ geometry assumed in design. Fabricated metal civil engineering shells are affected by circumferential welding, joint midsurface misalignment, cross-sectional ovalisation and plate thickness variations all of which introduce ‘imperfections’ that make the geometry deviate greatly from being that of a simple cylinder. Thus the characterisation of the shapes and extents of geometric imperfections that are likely to occur in full-scale fabricated cylinders for a particular application is of crucial importance to accurate design buckling strength predictions, computational design procedure development and construction quality assessment. Consequently, there is a rich history of laboratory-scale specimens being ‘scanned’ under controlled laboratory conditions to obtain careful imperfection profiles (1), although Sadowski *et al.* (2) recently demonstrated that such laboratory-scale specimens are frequently unrepresentative of full-scale metal shell structures and, if imperfection data meaningful to modern design practice is to be obtained, recommended that efforts should be redirected to scanning *full-scale* shell structures instead.

However full-scale shell structures are not surveyed as a matter of routine due to restrictions in site access, expense, technological challenges and data processing complexity. Where scans have been previously performed, the data processing methodologies have been inconsistent across research groups, frequently not presented in full and difficult to generalise to other applications of shells of revolution. Scanning full-scale shells presents additional difficulties which do not exist in the laboratory, not least the more challenging environmental considerations which should be accommodated in the scanning and processing methodologies, although datasets designed to control for this do not yet appear to exist. The Authors assert that there is a need for a *standardised* protocol for the repeated and reliable processing of high-fidelity laser scan point clouds of full-scale shells into gridded outer surface representations, or so-called ‘geometric digital twins’ (gDTs). Standardisation leads to automation, which in turn leads to a proliferation of advantages: more data, better data, better construction quality, better structural resistance assessments and longer structure lifetimes.

This paper develops a six-stage standardised software protocol for precisely this purpose. Building on the Authors’ experiences with laser scanning of full-scale towers and considerations from the mechanics of behaviour of metal shell structures, each progressive stage represents an independent step on the methods pipeline transforming a raw disparate point cloud into a clean meshable surface representing, ultimately, the load-bearing shell midsurface. Individual stages are justified based on

data quality and shell structural mechanics considerations with functional implementations presented in the MATLAB language and illustrated on a representative full-scale tower point cloud. The protocol is under active development and it is intended for improved methods and implementations to continuously replace existing stages. Limitations and best practices are also discussed.

2. History of shell imperfection surveys and the path towards modern gDTs

2.1 Reality capture of laboratory-scale shell specimens

The history of reality capture of shell structures is closely intertwined with scientific advances in the understanding of shell buckling phenomena. Given the strong sensitivity of the buckling capacity of shell systems such as cylinders under axial compression (3) to even minor deviations in the shell midsurface, numerous studies have tried to measure and assess these ‘imperfections’ and the reality capture of shell structures has thus been occurring in the form of ‘imperfection surveys’ since at least the late 1950s. Only a handful of studies are reported on in this paper, however a comprehensive review of imperfection surveys and measurement techniques from the 1950s until 1995 can be found in the works of Singer and Abramovich (4). One of the earliest attempts at measuring the real geometry of fabricated shells was likely by Wenk *et al.* (5) in 1954 who used a deflectometer to measure the radial displacements of circumferences of cylindrical specimens. Later in 1968, Arbocz and Babcock (6-8) used a ‘non-contacting pickup’ to measure the imperfect outer surface geometry of cylindrical test specimens. In 1977, Arbocz and Williams (9) investigated imperfections arising in aerospace shells by surveying an integrally-stiffened shell of 10 ft in diameter: measurements were taken by traversing a trolley supporting a direct current differential transformer (DCDT) along a guide rail on the outside of the shell. A ‘best-fit’ cylinder to the survey data was found using least squares formulations and net imperfections were computed by taking the difference in coordinates. Coefficients of a double Fourier series with half-waves (both sine and cosine) axially and full waves circumferentially characterising the imperfection field were determined and presented both graphically and in tabulated form. A Fourier series representation allows for significant data reduction of large datasets which was crucially important in the early surveys where computer memory was limited and expensive.

In 1978, Singer *et al.* (10) presented the results of 19 cylindrical stringer-stiffened shells surveyed using a linear variable differential transformer (LVDT) traversing a helical path inside a fixed specimen, with imperfections again presented as Fourier series coefficients. In 1979 Arbocz and Abramovich (1) established an ‘Initial Imperfection Data Bank’ (IIDB), a joint venture between Delft University of Technology (TU Delft) and the Technion of Haifa (4) to store the measurements of shells as double Fourier series coefficients (in a manner consistent with Arbocz and Williams

(9)) with the intention of housing the results of surveys of shells with different imperfections characteristic of different fabrication techniques to ultimately improve shell design (although the focus was always on aerospace shells). Five further reports (11-15) were released from TU Delft between 1988 and 1999 contributing to the IIDB on both laboratory-scale cylindrical shell specimens and the actual cylindrical / conical shells that formed part of the ARIANE launcher. Each report related to shells largely surveyed using LVDTs. The IIDB was pioneering in providing the first version of a general framework in which 3D imperfection data could be measured, processed and stored, but unfortunately no contributions have been made to it for some time and it is understood that the IIDB has been entirely defunct since at least the turn of the millennium.

Surveying technologies other than LVDTs have become increasingly employed. In 2010 Degenhardt *et al.* (16) surveyed ten carbon fibre reinforced plastic (CFRP) cylindrical shells using ATOS photogrammetry technology with a reported precision of 0.02 mm in measurements. The study aimed at enhancing the ‘knock-down’ factors applied to unstiffened cylindrical CFRP shells when computing buckling loads, with net imperfections again determined by the ‘best-fit cylinder’ method. Laser scanning techniques were by now becoming routinely deployed to survey laboratory-scale shells for their enhanced fidelity. For example, in 2018 Wang *et al.* (17) found that a finite element model with the real geometry incorporated was able to closely replicate the experimental result of an ‘integrally manufactured cylindrical shell’ of radius to thickness r/t ratio of ~ 333 under axial compression. The specimen had been surveyed using a Handyscan3D EXAscan laser scanner and specimen measurements had a reported accuracy of within 0.05 mm. Partial scans were taken around the cylinder and registered using targets glued to the specimen, with net imperfections computed relative to a perfect cylinder.

In 2024, Li *et al.* (18) created a 3D laser scanning system using an Arduino Uno, stepper motors and laser distance sensors to generate point clouds of specimens such as cylindrical shells, beams and welded joints, for the purposes of stability and fatigue assessment. Cylindrical shell specimens specifically were scanned in their entirety by rotating the specimens eliminating the need for registration of multiple scans. The method of Castro *et al.* (19), presented in more detail in Section 5.6 herein, was applied to generate finite element models using the ‘real’ measured geometries, although points deemed as not belonging to the shell were removed manually.

In 2024, Lin *et al.* (20) and Peng *et al.* (21) reported on the results of bending tests and surveys of cylindrical shells of 1 m in diameter and r/t ratios of 79, 106 and 158. Specimens were quarter-scale models of WTST sections fabricated by circumferentially welding five cans and were laser scanned prior to testing from six locations and registered in Artec Studio using cube targets placed on the specimen which served as reference points (20). The specimens were also scanned during

and post loading. The nominal geometry of the tubes was used as a reference geometry and registered with the point cloud to compute net imperfections. Cube targets were removed from the dataset and the resulting gap was filled by averaging local data points. Additionally, the weld beads were also excluded. The data was smoothed to reduce noise using a ‘non-local mean filter’ and a ‘box filter’ and the laser scan data was used to assess the out-of-roundness, eccentricity and dimple tolerances and classify the specimens according to EN 1993-1-6 (2007) (22). For the purposes of finite element modelling, the gap arising from weld bead removal was filled by fitting a Fourier series (21). To reconstruct the real geometry in finite element software (e.g. ABAQUS and ANSYS), nodal coordinates were shifted radially to match the closest point in the point cloud. Some discrepancies arose between the achieved buckling capacity in testing and the numerical simulations using the real geometries which the authors (21) attributed to the inpainting method perhaps not reflecting the true shape of a weld depression arising from welding. More recently, Ding *et al.* (84) reported on an experimental campaign involving 48 thin-walled steel cylinders subjected to combined bending and torsion, with each specimen laser-scanned before testing. Peng *et al.* (85) subsequently used the measured geometries of nine can-welded cylinders to validate detailed collapse simulations under pure bending. The associated open dataset, including original and processed scanned imperfections, material coupon results and bending test results, was released by Lin *et al.* (86).

Lastly, research into Wire Arc Additive Manufacturing (WAAM) structures is now also becoming more common and reality capture is routinely employed to assess the imperfections arising in the geometry and for the purposes of finite element modelling (23). For example, in 2024, Silvestru (24) released datasets from surveys of WAAM bar specimens and provided scripts for transforming scanning output into high quality meshes for finite element simulation.

2.2 Reality capture of full-scale shell structures

Full-scale shell structures such as aerospace launchers (25), silos (26-30), tanks (31-40), cooling towers (41,42) and penstocks (34) have been surveyed since the 1960s. A non-exhaustive review of the various field scanning campaigns of large-scale in-service shell structures is presented here. A focus is placed on the scanning and data processing methodologies to highlight steps shared between them and the need for a consistent geometric digital twinning protocol of axisymmetric shell structures for future studies. The surveys of full-scale spiral-welded tubes (43-46) are also

discussed herein as these appear to be at the limit of what can be tested in a structures laboratory, and to highlight the differences between laboratory-based and field-based scanning.

To the best of the Authors' knowledge, the earliest publication on surveys of in-service shell structures dates to 1964 where Lotmar (31) measured 54 cylindrical oil tanks 10.7-54.9 m in diameter and 9.1-16.1 m in height and generated radial profiles. Many methods including using a magnetically attached trolley and plumb line, theodolites and a Reduction Tachymeter were trialled. The preferred method of Lotmar involved a magnetic sled pulled up the tank with a ruler perpendicular to the shell wall. A pentagonal prism magnetically secured to the shell was used to read the measurement on the ruler. This also appears to be the largest campaign of surveys of in-service shell structures to date.

The earliest use of photogrammetry to measure shell structures appears to be from Shmutter and Ethrog (32) who in 1971 reported on the use of a Wild phototheodolite for cylindrical tank calibration. The circumferences of a 25 m radius tank were measured from individual photographs taken from 9 circumferentially located stations 40 m from the tank. A 'special calibration tape' was used to measure the circumference of the tank at desired locations to serve as control and to estimate the accuracy of the methodology (a mean square error of 10 mm in a circumference was reported). Following a cooling tower collapse which was attributed to imperfections of the shell, Chisholm (41) used stereo photogrammetry to survey three cooling towers 63-155 m tall and 59-85 m in basal diameter between 1975 and 1976 to generate stereoscopic models from overlapping images and subsequently assess net imperfections relative to the design geometry. Papo and Shmutter (33) later applied a similar method to calibrate a 17.63 m diameter cylindrical tank from the inside in 1978. In 1981 Faig (34) used stereo photogrammetry to assess aluminium alloy subassemblies of LNG spherical tanks and computed net imperfections from a best-fit sphere obtained via a least squares approach. Faig (34) also used mono photogrammetry to internally photograph and assess the diameter of a steel penstock. The best-fit circle of the cross-section was determined to obtain the cross-section diameter and assess the variation of the radii. In 1986, Ethrog (35) assessed the storm wind damage to a 14.62 m radius cylindrical tank using single photographs taken from outside the tank. 36 vertical profiles around the tank were measured with a vertical spacing of 1 m to determine the net imperfections from an ideal cylinder.

The earliest scan of a full-scale aerospace shell structure was possibly conducted in 1981 by Sebek (25) and was of the cylindrical and conical interstage shells of the ARIANE launcher, documented in parts III-V of the IIDB (12-14). The shells were scanned on a turntable using a LVDT (an axial scan with 10 mm between measurements was taken before the shell was rotated and scanned again) moving along a reference beam which was itself scanned to correct measured values of the

imperfections. A best-fitting cone or cylinder was computed, and the net imperfections determined before a Fourier series decomposition was performed. The shells were measured at the factory using a portable system from TU Delft.

The 1990 ‘Guide to the data reduction of imperfection surveys on circular shells’ by Cartalas, van der Hoeven and Klompé (47) explained and presented FORTRAN code (available only in printed format) which was prepared for processing imperfection measurements taken using LVDTs of the ARIANE launcher (14). The code first prepared the measurements for processing, with multiple corrections applied to the data such as linear interpolation of missing scans around the shell and reduction of the effects of reinforcement by adjusting measurements based on its thickness (14,47). A best-fit cylinder is then found, Fourier coefficients computed and some additional post processing of results such as plotting performed. The code outputs the net imperfections of the shell relative to the best-fit cylinder as well as the corresponding Fourier coefficients. This appears to be one of the first attempts at automating the processing of imperfection surveys of shells into a standardised manner. However, this particular codebase is understood to be unable to tackle the modern challenges arising from on-site laser scanning campaigns of full-scale, in-service, shell structures such as point cloud registration, can segmentation etc., and FORTRAN is now less widely used in data processing than it was before.

In 1988, Clarke and Rotter (26) reported on a survey of a 11.6 m diameter and 28.3 m tall, 2200 tonne capacity silo located at Glebe Island, Sydney. They aimed to devise a system of measuring imperfections in operational silos and tanks which was cheaper than alternative methods. The system involved a trolley with magnetic wheels pulled up the shell wall and measurements were taken at equal intervals of either 66.67 mm or 133.33 mm using an LVDT with a reported accuracy of 0.1 mm. A theodolite was used to relate vertical profiles and a ‘Krautkramer Ultrasonic Digital Wall Thickness Meter’ was used to measure plate thicknesses. The proposed data acquisition and processing (using harmonic analysis) methodology was verified using a remote surveying system (RMS2000) which makes use of theodolites and defines 3D coordinates with an accuracy of approximately 0.1 mm. A good match was obtained between the two systems, particularly for localised imperfections deemed to be critical.

Three steel grain silos at Port Kembla, Australia, were also surveyed in 1992 (27-30). Surface measurements were taken by a trolley supporting nine LVDTs and an electronic data logger pulled along the shell wall (28). Measurements were taken every 50 mm along a meridian (which is defined as the curve rotated about an axis to generate the axisymmetric shell) and every 500 mm circumferentially (30). A best-fit cylinder was found and net imperfections computed.

Measurements taken on the weld material appeared as outliers and were removed by cubic interpolation. The data was ultimately used for both 1D and 2D Fourier series decomposition to relate the fabrication process to the arising imperfection field. Unfortunately, the entire surface was unable to be surveyed leading to gaps in the data which interfere with Fourier decomposition, addressed to some extent by applying linear interpolation over the length of gaps to remove them.

In 2002, four thin steel tanks between 10 and 70 m in diameter were subjected to buckling tests under negative internal pressure (36). Three of the tanks had r/t between 800 and 3300 and were surveyed from the inside using a total station placed ‘as close as possible’ to the tank centre. Only cross-sections of the tanks at certain elevations were scanned, with 5 elevations for one tank (at a circumferential spacing of approximately 1890 mm) and 16 for the other two (at a circumferential spacing of 900 mm). Measurements had a reported accuracy varying between 2-3.5 mm. Additionally the centre of each cross-section was found by minimising the difference between radial measurements and the mean radius. Buckling-relevant fabrication tolerances (presented in Section 6) including dimple, unintended eccentricity and out-of-roundness tolerances were assessed to DIN 18800 (1990). Meridional and circumferential dimple tolerance measurements were made by physically placing a straight gauge at plate centres and across circumferential and meridional welds. Diameter values were measured by considering points on opposing sides of the shell to measure the out-of-roundness, while joint eccentricities were measured by considering points 5 mm on either side of welds using a dial gauge. A 2D Fourier series decomposition of the net imperfections from an ideal cylinder was also performed. In 2013, a steel tank which buckled due to a reduction in internal pressure was surveyed via terrestrial laser scanning using a Leica ScanStation C10 to assess the damage (37,38). The internal tank surface was scanned from two positions resulting in a point cloud of approximately 54 million points. Information from the top and bottom of the survey was used to fit a reference cylinder to the tank as these points were not expected to have undergone notable deformation. The difference between the reference cylinder and the measured cylinder was used to determine the tank deformations.

Two notable campaigns of laboratory-based surveys of full-scale spiral-welded steel tubes were published in 2015 (43) and 2016-2020 (44-46). The first survey relates to the RFCS Combitube project (43) where the outer surface of 18 cylindrical carbon steel tubes (16 spiral welded and 2 longitudinally welded) was scanned at TU Delft and the Karlsruhe Institute of Technology (KIT) before testing to failure under bending. At TU Delft, 15 specimens were placed on stiff supports in a horizontal orientation and inspected using the Sensopart FT50 RLA-40-F laser scanner traversing below the specimens on a trolley with a resolution of one measurement per millimetre of longitudinal axis, with specimens being rotated by 22.5° to 45° before being re-scanned. The

measurements from the scanner were related to a flat surface provided by a trough of liquid adjacent to the specimens to ensure a horizontal reference surface. The position of the trolley along the specimen was tracked using the rotation of one of the wheels. Each meridian was passed through a floating average filter to provide a smooth output. At KIT a different methodology was employed, with specimens placed on stiff supports in a horizontal orientation and scanned from 5 m away using a Leica ScanStation C10. Each specimen was rotated by 90° three times and surveyed as the scanner could only measure a quarter of the surface at once. Reference points were used by a third-party to register the surveys yielding point clouds with measurements approximately every 1.7 mm meridionally and circumferentially. A best-fit cylinder was fitted to the data and the net imperfections extracted then projected onto a regular grid using the MATLAB *griddata* function. The data was ultimately subjected to Fourier decomposition to analyse the imperfections induced by the spiral welding and approximately assess buckling-relevant fabrication tolerances (dimple, out-of-roundness and unintended eccentricity) of EN 1993-1-6 (2007) (22) using peak deviations and harmonic features.

The second study (44-46) considered eight steel tapered spirally-welded tubes 0.7 to 1.1 m in diameter ($100 \leq r/t \leq 175$). The specimens were supported by clamps in a horizontal orientation before the outer surface of the specimens was scanned using the Micro-Epsilon scanControl 2600-100 before the application of bending loads. The scanner was situated above the specimen and the specimen rotated to allow the laser scanner to achieve complete circumferential coverage. Specimens were scanned every millimetre in the meridional direction and at a spacing of 0.002 radians circumferentially with a reported accuracy of ± 0.2 mm. The compressed face of the specimens was also scanned during loading (44). A best-fit cone was determined, and the net radial imperfections were computed. A modification to counter the deformation under self-weight was not performed as these deformations were deemed to be negligible (45). Features which were found to be narrower and taller than a weld bead were considered outliers and manually removed (45), and the meridional and circumferential dimple tolerances were assessed (44) according to EN 1993-1-6 (2007) (22). The scan data was used to predict the location of buckling and capacity of the tested tubes using a proposed imperfection measure (45) and to assess the weld depression shapes characteristic of the specimens (46).

In 2020, Truong-Hong *et al.* (39) presented a method using ‘RANSAC-based voxelization’ to extract the tank wall from a larger point cloud consisting of other objects and illustrated the method on two tanks of diameters 18.3 m and 21.3 m respectively, scanned using a Faro Focus X130. The point clouds were registered and downsampled before the new methodology was applied. A best-fit vertical deformation and roundness assessment was also performed, likely based on

recommendations from the American Petroleum Institute (2014). Lastly, Truong-Hong *et al* surveyed a steel tank in 2021 (40) and created what they explicitly called a ‘digital twin’ of its geometry, possibly the first use of this terminology in the context of shell structures. A methodology of processing the data illustrated on a 16.5 m diameter tank was presented, and can largely be divided into two distinct sections, tank component segmentation and model creation. The shell wall, tank roof and floor were isolated and a series of fitted circles generated a finite element model for subsequent analysis using ANSYS. A ‘best-fit vertical deformation and roundness analysis’ was also performed based on guidelines of the American Petroleum Institute (2014). Whilst a core methodology was presented, it was not extended to recommendations regarding scanning procedure, point cloud registration, weld segmentation and point cloud inpainting. Additionally, fitted circles are unlikely to accurately represent asymmetric imperfections in the cross-section of a shell such as ovalisation and dimples, as a result of which high-fidelity models for accurate structural assessment of imperfection-sensitive shell structures are unlikely to be achieved. Most recently, in 2024, Makuch (42) reported on laser scan surveys of a 65.15 m tall cooling tower in relation to geometric imperfections. An in-depth review of surveys of cooling towers since 1973 is also detailed by Makuch (42) and is not repeated in this paper, however it is evident that numerous studies have measured geometric imperfections relative to a nominal model of the shell.

2.3 Geometric digital twinning

The term ‘geometric digital twin’ (gDT) has recently been adopted to describe a digital representation of the ‘as constructed’ geometry of a structure (48-54), proposed as a means for structural health monitoring (48-50) and advanced realistic structural analysis (51,52). The term ‘as-built building information model’ carries a similar meaning and has also been used to relate to such models (55-57). The generation of gDTs typically requires point clouds obtained from laser scanning (48-50,52,54), although photographs have also been used (51), and methodologies for developing gDTs of bridges (48,54), pipelines (49), structural members of buildings (50) and masonry (51,52) have recently been developed. The Authors have similarly chosen to adopt the term ‘gDT’ herein, with the strict meaning that it represents a high-fidelity geometric model of a nominally axisymmetric thin-walled metal shell structure.

Established methods of generating gDTs typically make use of prior knowledge of the structure in question and tailor methods accordingly. For example, Li *et al.* (49) utilised the cylindrical nature of pipelines to collapse the 3D point cloud into multiple 2D circular slices and applied a polar coordinate transformation to divide the slices into cells circumferentially for further processing. Mirzaei *et al.* (50) also made use of prior knowledge about structural design to attribute data points

to members, for example by considering the congestion of beams near floors and normal vector orientations of columns. In keeping with gDT generation methods presented for other structural forms (48-54), a general protocol for processing axisymmetric shell structures should likewise exploit the axisymmetric nature of the structure and be specialised to exploit their characteristic features arising from the manufacturing process such as weld beads. gDTs were also created to varying levels of fidelity depending on their intended use. For example, when creating gDTs of three-leaf stone masonry walls, Saloustros *et al.* (52) laser scanned each individual stone before wall construction to accurately capture the wall's microstructure as this is necessary to validate the existing numerical modelling techniques applied to stone masonry. Similarly, the reconstructed geometry of an axisymmetric shell should be of sufficiently high quality to ensure that all structurally-relevant features are accurately captured.

The Authors acknowledge that digital twinning is not limited to the creation of high-fidelity model representations, as the complete paradigm must incorporate bidirectional feedback from live sensors to continually forecast the state of the system (e.g.(58-61)). The feedback dimension appears to be currently achieved only through discrete point-wise sensors (strain gauges, transducers, accelerometers) coupled with relatively simple finite element representations of the structure treated as a beam (e.g. (62,63)), such that the realism of the true structural form or the associated complexity of response is not currently captured. The complete framework for creating gDTs of shells of revolution presented in this paper aims to facilitate the path towards the holy grail of spatially and temporally continuous reality capture providing complete-system real-time feedback for a true complete digital twin of this class of structure.

2.4 A summary of the historical trends and extrapolation to future gDTs

A number of historical trends may be identified based on the detailed literature review presented above. First, the initial step of geometric data acquisition has evidently experienced a major evolution over time. The earliest attempts of geometry capture on small laboratory specimens (10,11), the ARIANE launcher (12-14,25) and the silos of Glebe Island (26) and Port Kembla (27-30) measured the surface profile through direct contact with LVDTs, but there has been a clear shift towards long-distance measurements using non-contact laser scanning (36-40,42-46) and this appears to be deployed ever more frequently on full-scale shell structures. Photogrammetry (32-35,41) was used in the 1970s-1980s, but does not appear to be routinely employed for shell surveys at this time. The trolley/sled systems used by Lotmar (31), at Glebe Island (26) and Port Kembla (27-30) and the reference beam used for the ARIANE launcher (12-14,25) would be difficult to adapt to all field conditions and construction types (e.g. stiffened or corrugated shells), but laser

scanning offers a mobile technology that is much more adaptable to site conditions and can readily be employed in the field. Thus, any methodologies developed to automate gDT creation for the high-fidelity modelling of axisymmetric shell structures should be applicable in the first instance to point clouds generated by laser scanning (or an equivalent optical method).

Second, the ARIANE launcher (12-14,25), the large spiral welded tubes of (44-46) and some of the tubes of (43) relied upon the ability to rotate the shells and scan individual meridians allowing for almost full circumferential coverage with a desired uniform density. Even very recent scanning programmes continue to rely on physically rotating the shell specimen, such as the system of Li *et al.* (18). Since only a single scanning position is used, the (potentially very complex) task of correctly combining multiple scans (registration) is mitigated. However, rotating shells or the reality capture system about a horizontal or vertical longitudinal axis is only possible in a laboratory setting and is currently not deployable to vertically freestanding installed in-service shell structures which are the focus of this paper (various drone-based proposals are understood to be only at the concept stage at this time). Hence, any general methodology must explicitly handle point cloud registration.

Third, Fourier series decomposition was employed routinely in the processing of data from scans of both laboratory and in-service shells (1,6-15,17,26,30,36,43,46) mostly for data reduction and subsequent structural midsurface reconstruction, a method born in an era when computer memory was at a premium. Unless Fourier harmonics are explicitly desired for data exploration, for example for the characterisation of wave-like imperfection features, the method is no longer a necessary core of the data processing methodology in an era of abundant computer memory nor is it desirable given that it has always been sub-optimal in handling irregularly-spaced data (64), irregular boundary conditions, missing regions or systematic outliers.

Fourth, many of the afore-mentioned studies of laboratory and in-service shells adopted some variant of a ‘best-fit’ (and geometrically perfect) geometry to create a reference entity relative to which ‘net imperfections’ can be established for the purposes either of structural surface reconstruction or buckling-relevant fabrication tolerance assessment. As will be demonstrated in this paper, while a ‘best-fit cylinder’ is helpful in reconstructing a load-bearing midsurface it shouldn’t be necessary to establish what is and is not an imperfection for the purposes of a geometric tolerance assessment, and is not necessary if using the EN 1993-1-6 (2025) (65)

methodology whose system of ‘gauges’ is designed such that each one acts as an artificial local reference axis relative to which tolerances may be measured.

Lastly, as data acquisition and processing techniques have developed, data quantities have grown from handfuls of Fourier coefficients which can fit on a printed page to three-dimensional GB-sized data clouds consisting of hundreds of millions of coordinates. The creation of gDTs is thus rapidly becoming a problem requiring the deployment of HPC-grade computational resources, and awkward laborious manual processing (e.g. to remove outliers) must in future be replaced by automated solutions based on machine learning methods. A standardised modular software protocol for this purpose would provide a valuable framework within which methods and code can be incrementally developed and improved, and presenting one is the purpose of this paper.

3. Aims and scope

This paper aims to automate the high-quality geometric digital twinning of metal shells of revolution for fabrication quality assessment, imperfection characterisation, finite element simulations of as-built geometries, structural health assessment and lifespan extension. A multi-stage standardised protocol for the automated processing of laser scan point cloud data into a high-fidelity gDT representation of in-service, full-scale, axisymmetric shells is developed. The output of the protocol is a gridded surface representation of the shell which can be readily used for feature exploration, fabrication quality assessment and finite element model development of geometrically realistic shells. The protocol serves as a novel geometric digital twinning framework, with working methods presented for each step that can readily be interchanged with improved alternatives if desired or once these become available. This protocol and its constituent algorithms are presented and have been tested in the context of a recent high-definition laser scan of a real full-scale WTST. Given the growing interest in controlling the fabrication quality of constructed shells, novel algorithms are additionally presented for assessing the buckling-relevant fabrication tolerances according to EN 1993-1-6 (2025) (65) allowing consistent and efficient tolerance verification. A working MATLAB implementation of each stage of the protocol and tolerance assessment algorithms are disseminated via GitHub.

4. Unique laser-scanned WTST testing and development dataset

The geometric digital twinning protocol presented shortly will be illustrated on a unique GB-sized dataset of a laser-scanned full-scale offshore WTST gathered specifically for the purposes of its development and released as part of this publication. The steel tower was fabricated by circumferentially welding multiple cylindrical and conical shells known as ‘cans’ together to form sections which are subsequently connected by bolting together flanges at the end of sections. Being at the final stage of tower delivery it was fully assembled with all sections joined together ready

for shipping (although without the nacelle), and the last opportunity where an offshore WTST can be scanned on land using a laser scanner and with geometric imperfections closest to what would be present in the final offshore structure. The painted outer surface of the WTST was measured and it is assumed that this surface is representative of the underlying unpainted metal outer surface. The tower was located adjacent to others and scanned in an upright position using a Leica ScanStation P40 from several positions at ground level. The chosen scanner offers a high scan rate of up to one million points per second and a root mean square range noise of 0.5 mm at 50 m, with the recommendations of Kathirkamanathan *et al.* (66) regarding the minimum scanning resolution of WTSTs adhered to. It is important to note that this minimum recommended resolution is a lower bound if the scanned points were directly used to create a mesh representation. Algorithms such as the ones required by the protocol require a far greater number of points for local plane fitting, correct identification of outliers and for weighted averages of data points. The maximum resolution of the scanner used easily satisfied this requirement.

The tower was scanned from multiple positions around the circumference, governed by the arrangement of the towers on site. Scanning was performed on a day with a clear forecast where the wind speed was expected to be low (< 10 km/h) resulting in reduced movement at the top of the tower. Initial scan registration was performed by the scanning contractor using physical control points placed around the tower and commercial software. Upon examination of the data, the contractor was asked to take additional measures to improve registration quality. The contractor first used manually created control points in overlapping regions of the scans to improve the registration. Cloud-to-cloud registration was then used which considers the points in the clouds themselves instead of relying upon physical or simulated control points. Only the scans from positions which were close to the tower were included in the dataset. The scanning contractor additionally manually removed certain features from the dataset which did not relate to the shell such as the ground and large obstructions such as staircases, although it is also possible to remove these automatically.

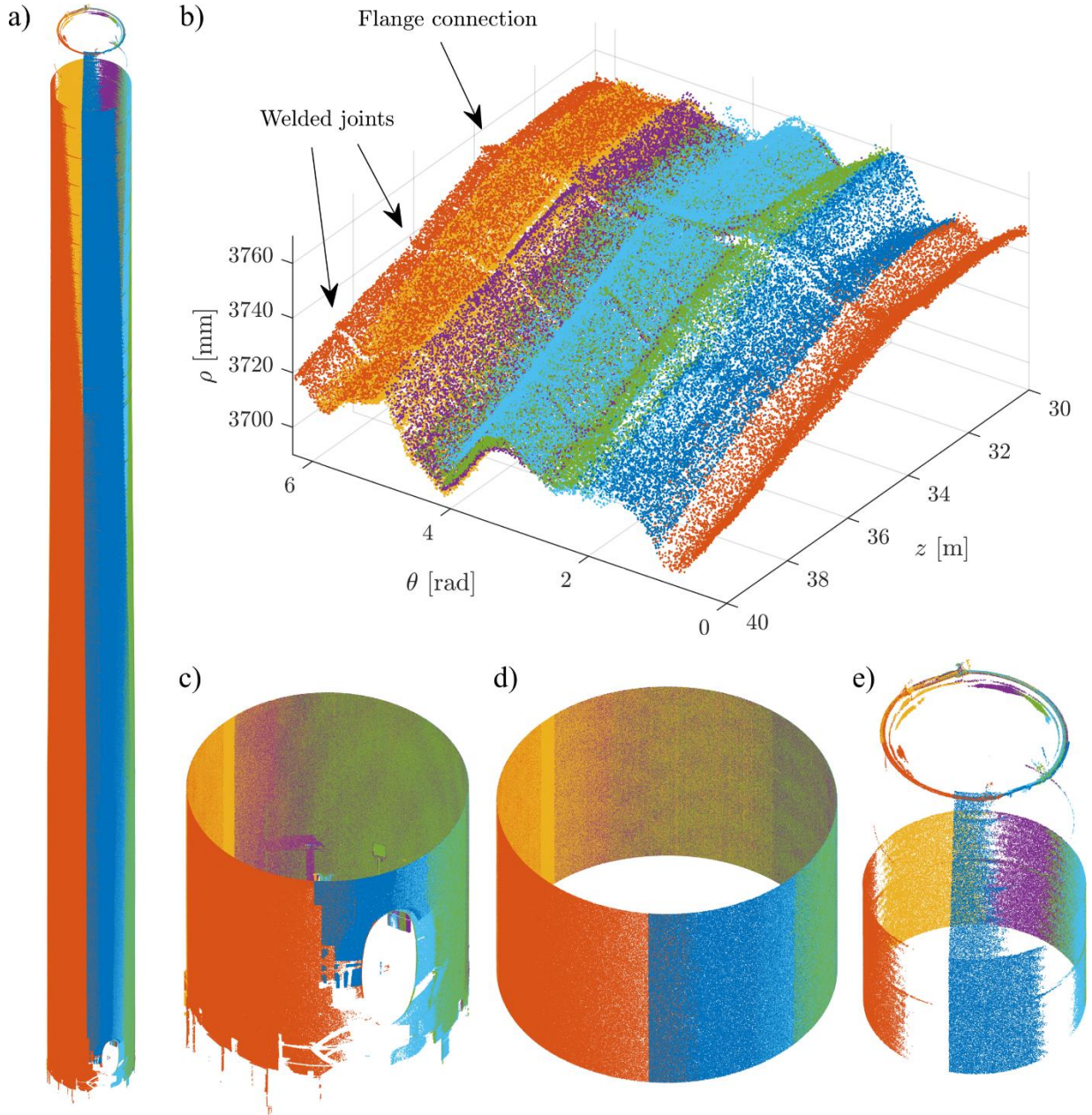


Figure 1: WTST dataset used for the geometric digital twinning protocol development: Full tower in a) Cartesian and partial tower region in b) cylindrical coordinates, with c) base feature including door, d) midheight and e) top boundary. Different colours represent data from scans made from different positions.

Along with sample MATLAB implementations of the various stages of the protocol, a point cloud dataset is provided as a sandbox for users to test the algorithms presented. The provided dataset has been augmented through scaling by different constant factors in the radial and meridional directions such that it can no longer be assumed to represent any specific existing tower. Users should therefore not use the dataset to draw any conclusions about the construction quality of the tower itself and only use it for algorithm testing and development. All illustrations in this paper relate to the augmented dataset. An example of the augmented point cloud in Cartesian (x, y, z) and cylindrical (ρ, θ, z) coordinates is presented in Figure 1a) and b) respectively (with important

imperfection features such as welds and flange-to-flange joints highlighted). The figure shows point clouds of different colours where each colour relates to a different scanning position and thus a different cloud registered by the scanning contractor. Figure 1c) shows the bottom of the tower where there are gaps in the data due to an opening for a door, but also due to obstructions caused by attachments to the tower such as staircases. Figure 1d) shows the middle portion of the tower which has good coverage and will be the focus of the methods presented in this paper. Finally, Figure 1e) shows the top of the tower where degrading levels of coverage vertically and circumferentially are visible.

5. The high-fidelity geometric digital twinning protocol

5.1 Introduction

A six-stage automated protocol has been developed to convert point clouds from multiple scan positions into a high-fidelity gDT with a gridded outer surface representation on which fabrication tolerance assessments and realistic finite element modelling can be performed. The protocol is currently specialised to full-scale, in-service, cylindrical and steeply conical shells fabricated with circumferential and meridional plate welding surveyed in the field as opposed to under laboratory conditions. This methodology has been tested and illustrated on a WTST but is readily generalisable to other axisymmetric shells such as silos and tanks. Each stage of the protocol is to be applied sequentially but certain stages can be bypassed depending on the circumstances explained in the relevant sections. An additional Stage 0 is also included which relates to recommendations to the scanning contractor aiming to maximise the quality of data entering Stage 1 proper.

5.2 Stage 0: Data acquisition and recommendations to the scanning contractor

Providing the best quality point cloud input into the protocol is vital for a high-fidelity shell outer surface representation. The factors which can affect the quality of individual scans include the choice of scanner and applied settings, environmental conditions (temperature and lighting), surface properties (surface material roughness and reflectivity) and scanning geometry (location of scanners relative to target surfaces) (67). The scanner location controls the angle of incidence of the laser (angle enclosed between surface normal and line of sight of laser) and has been shown to impact scan quality. For example, when considering a Faro 880 scanner, Lichti (68) found limiting the maximum angle of incidence to 65° would reduce outliers in the dataset. Controlling the environmental conditions and the properties of the prospective scanned surfaces is difficult on site, though the choice of scanner locations can be controlled to some extent if conditions on site allow.

The following are the Authors' recommendations based on recent experiences and existing literature to scanning contractors when surveying axisymmetric shell structures to help achieve the best input into (and thus output from) the geometric digital twinning protocol:

- To survey shells, use a high-resolution laser scanner whose scanning accuracy is small compared to the length scale $\sqrt{rt}$ for short-wave shell bending.
- Where possible, survey the shell from the inside from a single point to prevent the need for registration and the corresponding errors. If the shell cannot be scanned internally, it is strongly encouraged to ensure large overlaps between scans from different positions as this will vastly improve the quality of registration performed on the dataset. Scanning coverage degrades with height and thus ensuring large overlaps is particularly important to ensure gaps do not arise in the dataset. Site conditions should be arranged to accommodate this as far as is possible.
- Determine beforehand the maximum angle of incidence to ensure sufficient scan positions are used and that there is reduced noise and random outliers in the point clouds.
- Survey as much of the shell as possible to avoid the need for gap inpainting in software.
- Survey the shell as late as possible in its fabrication and installation process to have the most representative structural features that would be present when the shell is in operation.
- Survey the shell in its free-standing vertical configuration as opposed to on its side to prevent the need for a correction for the sag of the shell under its self-weight.
- Retain any RGB (Red-Green-Blue) or surface normal data from the surveying as it can be used for colour-based outlier removal.
- Plan surveying for days where there are low wind speeds to mitigate any sway of the structure which can introduce noise in the data and make smaller details difficult to discern.

The data provided by the scanning contractor should be just the structural shell with other features such as the surrounding environment and non-structural components including staircases or pipes etc. removed. Each individual scan after registration should be provided such that the end user can check the quality of registration and make improvements, if necessary, as described in Section 5.3. The centroidal axis of the scanned shell should be adjusted by the contractor to lie upon the origin and the full dataset should be provided with *no* downsampling or floating-point rounding at this stage. The vertical coordinates of the point cloud should be adjusted such that they coincide with the expected agreed-upon nominal elevations (for example, the base of the shell could be positioned at a vertical height of 0 m). The GitHub repository additionally provides an example records template which should be adapted for the specific site and ideally completed by the contractor on each day of scanning.

5.3 Stage 1: Enhanced registration

Laser-scanned point clouds of tall or long axisymmetric shell structures may appear well-registered in Cartesian coordinates after being passed through the relevant software functionality, but a conversion into a cylindrical coordinate system may reveal significant misalignments in the scans. This is because while the mm-scale misalignments may be negligible relative to the m-scale height or diameter dimensions of the structure, they are non-negligible at the mm-scale of the wall thickness. An example of such a misalignment present even after very careful registration by the scanning contractor can be seen in Figure 2a) which shows a cross-section of the WTST in cylindrical coordinates. Whilst the registration quality achieved by the contractor is likely sufficient for non-structural applications (and errors like these would likely be insignificant if encountered for many other structural forms), the sensitivity of shell structures to mm-scale imperfections in the midsurface geometry can lead to very incorrect structural assessments if artificial misalignments like this are permitted to remain in the dataset. To avoid this, further ‘enhanced’ registration is almost certainly required for generating genuine gDTs of axisymmetric metal shells.

In the present implementation, the point clouds were downsampled prior to further registration to reduce computation times. Since a cylindrical coordinate system is employed in this stage of the protocol, a best-fitting truncated cone was determined using least-squares to ensure that the shell is centred at the origin. The horizontal position (x_{bc}, y_{bc}) of the cone centroidal axis, its angles of rotation α and β about (x_{bc}, y_{bc}) and the bottom and top of shell radii r_b and r_t (i.e. radii at the lower and upper elevations of the analysed point cloud region) were solved for and the corresponding transformation applied. An example of a best-fitting cone procedure with similar transformation parameters can be found in (69). The vertical coordinate of the bottom of the shell is assumed to be correctly located at an elevation of 0 m by the scanning contractor.

The Iterative Closest Point (ICP) algorithm (70) is a widely adopted technique for point cloud registration, and consists of two iterative steps. The first is finding the correspondences between points of two different clouds using nearest neighbours. The second is subsequently determining a rigid transformation which minimises a chosen metric (71), frequently the Euclidean distance between points (70) though other metrics such as the point-to-plane metric (72) which accounts for the local surface normal are also commonly adopted. Additionally, Bergevin *et al.* (73) explain that when considering multiple scans, applying the ICP algorithm involves the registration of two scans at a time sequentially, gradually building up a unified model. However, once registration of a pair has been performed their relative positions are fixed and information from additional scans cannot improve the registration of already considered scans. Bergevin *et al.* (73) instead recommend registering all scan positions simultaneously.

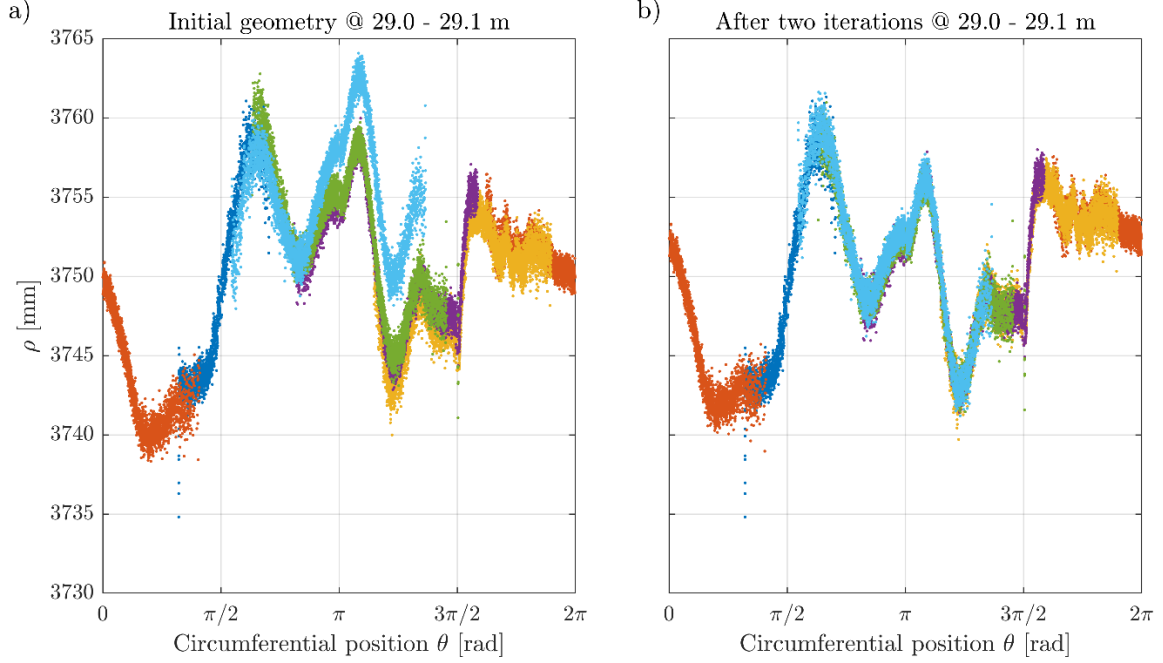


Figure 2: Illustration of Stage 1 enhanced point cloud registration.
Different colours represent distinct point clouds scanned from separate locations.

A modification of the ICP algorithm (with point-to-plane metric) tailored to axisymmetric shells to improve the registration achieved by the scanning contractor is presented in Algorithm 1 (a MATLAB implementation may be found in the GitHub repository). The correspondences between scans are obtained via an exact nearest-neighbour search using a k-d tree data structure. Only nearest neighbours closer than a cutoff distance (20 mm was adopted) are considered in the registration to avoid correspondences between distant points, and working in cylindrical coordinates aided efficiency. This is because circumferential binning in cylindrical coordinates reduces the number of candidate point pairs, while exact nearest-neighbour distances are evaluated in Cartesian space. The points of each scan position are first divided according to their circumferential coordinate into 100 bins spanning from 0 to 2π . This is illustrated in Figure 3a) on a vertical slice of height 0.1 m of the data of two of the scans presented in Figure 2a). Bins in which points exist for each scan are identified in Figure 3b) and c) where they are marked with the same colour as the scan. For a pair of scans, a nearest-neighbour search is only performed for a point of one cloud if it is in a bin shared by the other cloud to reduce computation time. The maximum circumferential angle $d\theta$ between two points that are within the cutoff distance is approximated as the cutoff distance divided by the minimum shell radius. The bins where points exist are extended by as many bins as necessary to account for $d\theta$, an example of which is marked on Figure 3b) and c) by grey boxes with crosses. This ensures potential correspondences between points of two scans that are closer than the cutoff distance but are in non-overlapping bins are not missed. Figure 3d)

displays the bins considered to overlap after circumferential extension for nearest-neighbour searches.

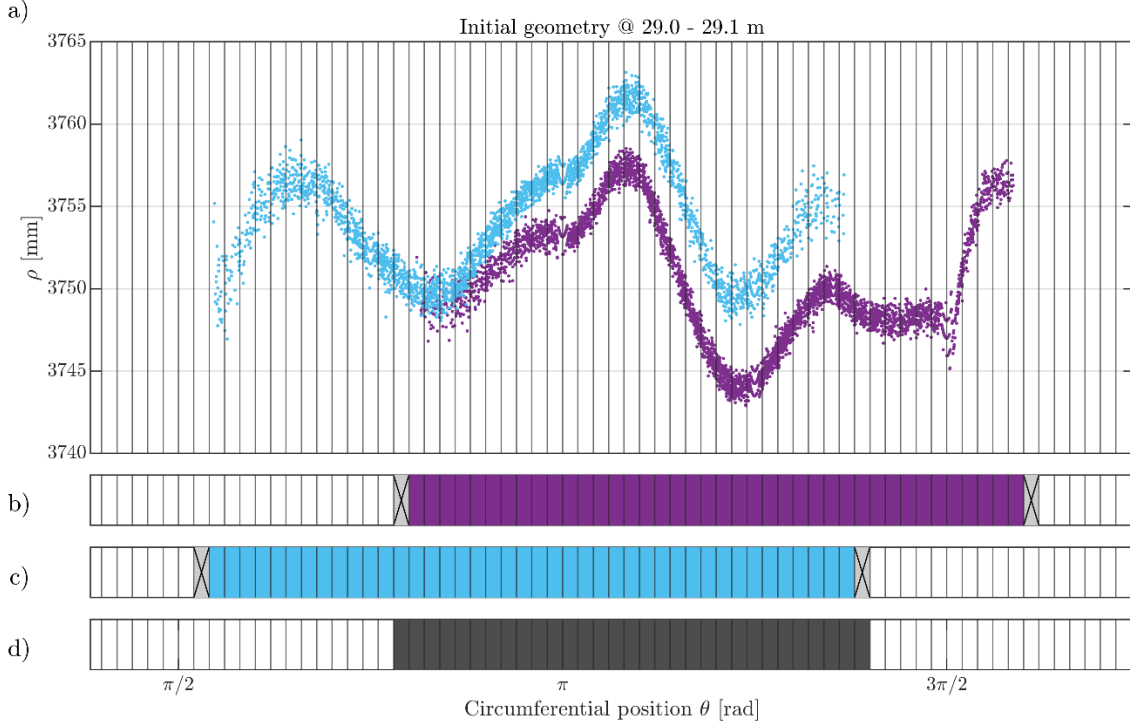


Figure 3: Circumferential binning of points to identify overlaps.

For each correspondence, the signed point-to-plane residual e_i (72) is defined in Equation (1). The rigid transformations are obtained by minimising the sum of squared residuals across all correspondences between clouds using the nonlinear least squares solver ‘lsqnonlin’ of MATLAB 2023a. Equation (1) is presented in terms of a pair with index i of points a_i and b_i belonging to clouds a and b , while T_a and T_b are the transformations applied to clouds a and b respectively and n_i is a local surface normal estimate at a_i . Each scan except one is free to translate and rotate in all six degrees of freedom: one scan is fixed to prevent a solution where there is a global translation or rotation of the entire structure.

$$p = \sum_i e_i^2 \text{ where } e_i = n_i \cdot (T_a \cdot a_i - T_b \cdot b_i) \quad (1)$$

Typically, only a single iteration was found to be necessary to obtain a significant improvement over the initial contractor-supplied alignment, however the algorithm has been implemented in an iterative manner in case subsequent iterations provide improvements for other datasets. A maximum of 5 iterations is likely to be more than sufficient. The registration after just two iterations is presented in Figure 2b) and illustrates a significant improvement over the initial contractor registration made with industry software. The registration quality is assessed by computing the 99th percentile of the point-to-plane metric (P_{99}) after each iteration using the updated correspondences

between scans. This metric typically reaches a plateau after a single iteration and does not necessarily continuously decrease with additional iterations. Thus, the iteration where the metric is lowest is chosen as the best.

Applying the present algorithm can lead to some vertical misalignment at joints which the user must be wary of. These misalignments likely arise due to the shell normal vectors pointing predominantly in the radial direction and thus not sufficiently constraining vertical movements when applying ICP with the point-to-plane metric. The user is advised to check which iteration has the best result considering a visual inspection of the alignment and P_{99} . Ensuring good joint alignment is important as Stage 2 of the protocol relies on the joints for can segmentation. Methods such as ICP work best when there are overlapping regions between scans, and alternate methods may be required if scanning conditions on site do not permit overlapping scans to be taken. The key inputs to Algorithm 1 are the maximum number of iterations (`numIter`), the cutoff distance for nearest-neighbour points (`cutoffReg`) and the individual point clouds (`clouds`). The outputs of the algorithm are the 2D array of registration parameters applied in each iteration where each column corresponds to the parameters of a single iteration (`regParams`), the best-fitting truncated cone parameters solved for at the start of the algorithm (`bestFitParamsPreReg`) and the iteration with the lowest P_{99} (`iUse`).

5.4 Stage 2: Can segmentation

5.4.1 Introduction

The segmentation of a shell into the individual cans is needed to assign thickness values to regions for finite element modelling and for the assessment of fabrication tolerances such as unintended eccentricity at joints. However, automated segmentation is not possible in the absence of nominal information (such as can heights from design drawings) or if there is a lack of confidence regarding plate length tolerances in can fabrication. A method of performing can segmentation is presented here assuming that the axisymmetric shell has been fabricated by circumferentially welding cans. Identification of systematic features such as weld beads which are present at every circumferentially welded joint and identifying the transition between flanges at flange-to-flange bolted connections allows for the joint region to be removed and the individual cans to be segmented. The weld beads at joints do not relate to the true shell surface and would adversely affect structural modelling and fabrication tolerance assessments if not removed.

A methodology for performing can segmentation is presented here, consisting of the following three steps: global radial profile generation, joint ‘region-of-interest’ (ROI) identification and finally joint segmentation, which returns the vertical coordinates of the top and bottom of each can which are just clear of the joints. This stage can be skipped entirely if the shell under consideration

645 consists of a single can with no circumferential joints. If the shell contains multiple cans which are
646 not circumferentially welded together, an alternate method may be required for automated can
647 segmentation.

Algorithm 1: Enhanced registration

Input: numIter (max. no. of iterations), cutoffReg (nearest-neighbour cutoff distance), clouds (point clouds to be registered)

Output: regParams, bestFitParamsPreReg, iUse

Compute number of data clouds, numClouds.

Fit a best-fitting truncated cone to all clouds at once, apply transformation and store parameters in bestFitParamsPreReg.

Initialise regParams to store registration parameters.

for iteration in (1 to numIter + 1) **do**

for cloud in clouds **do**

 Divide cloud into circumferential bins and identify if there are points in each bin.

 Extend bins circumferentially by necessary amount.

end for

for firstCloudIndex in (1 to numClouds) **do**

for secondCloudIndex in (firstCloudIndex + 1 to numClouds) **do**

for overlappingBins between firstCloudIndex and secondCloudIndex clouds **do**

for point in overlappingBin from cloud with secondCloudIndex **do**

 Find nearest neighbour of point, to points in cloud with firstCloudIndex.

end for

end for

 Remove nearest neighbours with distances more than cutoffReg.

 Store correspondences in array called connections.

end for

end for

 Compute point-to-plane metric for all correspondences and store 99th percentile.

if iteration equals (numIter+1) **then**

break

end if

 Solve ICP minimisation problem with point-to-plane metric using lsqnonlin for all clouds simultaneously and obtain transformation parameters.

 Apply solved registration transformation to clouds and append to regParams.

end for

Identify iUse, the iteration with smallest 99th percentile of the point-to-plane metric.

5.4.2 Global radial profile generation

The shape of circumferential weld beads is not circumferentially uniform due to the inherent variability in the welding process. When performing a coarse segmentation of butt weld beads to improve the segmentation accuracy of their proposed algorithm, Huang *et al.* (74) aggregated the global variation perpendicular to the direction of welding into a single global bead profile. The Random Sample Consensus (RANSAC) algorithm was then used to fit a quadratic polynomial to the data where points corresponding to the weld bead became outliers which could be identified. This methodology was applied after the approximate location of the weld bead had already been found. However, the method of generating a global bead profile can also be applied to the entire point cloud to generate a global radial profile for the entire shell where multiple weld beads are clearly visible and can each in turn be identified and then segmented. Using a global radial profile allows the key geometric features to be presented in only two dimensions as opposed to three, simplifying segmentation. An algorithm for generating the global radial profile is presented in Algorithm 2 and an example profile generated by the proposed methodology is presented in Figure 4 where the weld beads in the cylindrical region of the considered WTST are clearly visible. The transition between adjacent bolted flanges can also be clearly identified. The resulting global radial profile is only to be used for can segmentation and not for trying to identify the nominal properties of a surveyed shell.

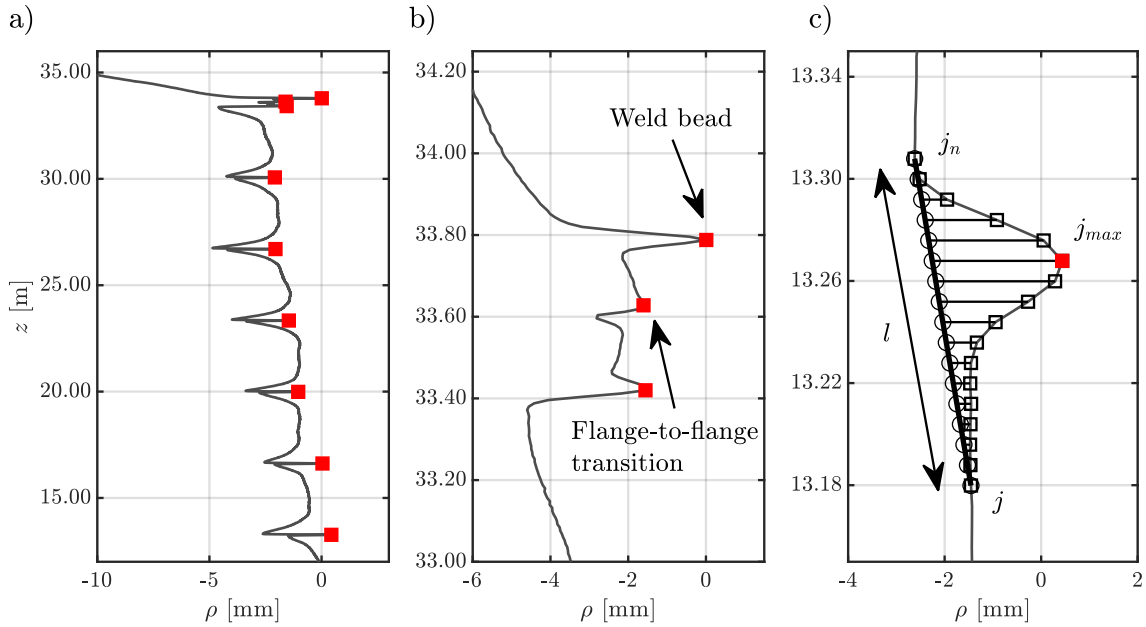


Figure 4: Illustration of Stage 2 global radial profile generation. a) Section of global radial profile with identified joints. b) Close up of flange-to-flange transition and weld bead. c) Joint ROI identification as per Algorithm 3.

Algorithm 2: Global radial profile generator

Input: ρ , θ , z (points of combined registered point clouds), b_w (expected bead width), n_{sub} (number of vertical sub intervals) and n_c (number of circumferential windows).

Output: globalRadialProfile defined by coordinates z_G and r_G

Compute best fitting cone and apply transformation to centre and orientate cone upright.
 Compute desired global radial profile vertical spacing p_s and vertical window height w_h .
 Define windowCentreZ, the vertical coordinates of the centres of the vertical windows using w_h and n_{sub} .

Define windowCentreTheta, the centre of the circumferential window using n_c .

Initialise z_G and r_G to store the global radial profile.

for vertical sliding window in windowCentreZ **do**

 Identify pointsVerticalWindow, the points which lie in the vertical window.
 Initialise arrays θ_c , z_c , r_c to store the circumferential, vertical and radial coordinates after radial correction.

for circumferential window in windowCentreTheta **do**

 Identify the points of pointsVerticalWindow which lie in the circumferential window.
 Subtract median radius of the circumferential window from the radial values inside it.
 Append the radially corrected coordinates to θ_c , z_c , r_c .

end for

Define arrays z_s and z_m containing the vertical coordinates of the bounds and centres of the n_{sub} sub intervals within the vertical window.

Initialise array r_m to store median values within a vertical window.

for vertical sub interval in vertical window **do**

 Identify points within vertical sub interval and store median of their r_c values in r_m .

end for

if first vertical window **then**

 Append z_m to z_G .
 Correct values in r_m so that the array starts at zero and then append to r_G .

else

 Correct values in r_m so that the array starts at last value of r_G .
 Remove the first value of z_m and r_m and append to z_G and r_G respectively.

end if

end for

671

672

673

674

675

A best-fit truncated cone is determined using a least-squares method as in Section 5.3 and the point cloud is considered in cylindrical coordinates. The target vertical point spacing p_s of the global radial profile is defined which relates to the size of the features to be captured by the profile i.e. weld beads and flange-to-flange transitions. Two sets of sliding windows are employed in the

algorithm, one vertical of height w_h , ($w_h = n_{sub} \cdot p_s$) and width 2π and the other circumferential of height w_h , and width $2\pi / n_c$. The parameter n_{sub} relates to the number of vertical sub-intervals the vertical window is divided into, while n_c relates to the number of circumferential windows. A vertical window is considered and points within the window are divided into n_c circumferential windows. The median radial value of each circumferential window is subtracted from radial values in the circumferential window to correct for circumferential variations. These median corrected values become the new data points within a vertical slice which is further divided into vertical sub-intervals. The median radial value of each sub-interval is computed and together define a portion of the global radial profile. The next vertical slice above the current slice is extracted and the process repeated until the portions of the global radial profile computed are stitched together to generate one complete profile. The sliding window centres are defined such that the profile can be stitched together by ensuring radial continuity at two points of the same vertical coordinate.

The point spacing p_s of the global radial profile should be as large as possible whilst representing the intended features as it leads to a smoother profile. In the context of the example WTST data considered here, the point spacing was related to the expected weld bead width b_w i.e. $p_s = b_w / 10$. The variable n_{sub} serves to scale the window size w_h , with larger values ($n_{sub} = 1000$) found to create a bumpy global radial profile ($n_{sub} = 10$ was finally adopted). Applying a circumferential median correction helps generate a smoother global radial profile than otherwise and as such $n_c = 100$ was used. If the weld beads do not appear similar to Figure 4 when applying this algorithm, issues may arise when applying the methods presented in Section 5.4.4 during joint segmentation. In such rare cases, the user is advised to apply the registration algorithm again.

5.4.3 Joint ‘region-of-interest’ ROI identification

The location of joints can be computed automatically using the global radial profile. An algorithm was developed to find the location of sharp variations in the global radial profile which arise due to weld beads (but can also arise at the interface between flanges as illustrated in Figure 4) and correspond to joints. Sharp variations can be identified by comparing the profile against a reference line of length l between two points on the global radial profile as shown in Figure 4c). Computing the difference between the profile and the reference line for all points along the profile provides a measure of the magnitude of the sharp variations in the profile which can be used to identify the joints. The location and magnitude of the maximum deviation between the profile and the reference line is stored for each line position. A reference line of length equal to 1.5 times the expected bead width b_w i.e. $l = 3 / 2 b_w$ was applied in the context of the example WTST dataset.

After the reference line has been applied to the entire global radial profile, a moving maximum filter is applied using the ‘movmax’ command of MATLAB to combine nearby positions of

deviations from the global radial profile. The ‘movmax’ command is applied with a window of width l , however, the window width must be smaller than the smallest distance between adjacent joints to prevent adjacent joints from being merged. In the case of WTSTs, the limiting value for the window width is the height of a flange and was satisfied by the adopted window width. The ‘islocalmax’ command is used to work out local maxima. Since the number of joints (`nJoints`) in a considered region is known in advance, the vertical coordinates of the `nJoints` largest local maxima must correspond to joints between cans. An example of the identified joints using this algorithm is presented in Figure 4. Additionally, as illustrated on Figure 4c), the weld bead has an approximately symmetric shape about its midline resulting in the peak deviation between the reference line and global radial profile occurring at its centre and thus an accurate joint centroid is identified. The flange-to-flange transition however is not symmetric as seen on Figure 4b) and thus a reduced accuracy in joint centre identification can be expected. The developed algorithm is summarised in Algorithm 3.

Algorithm 3: Joint ‘region-of-interest’ ROI identification

Input: z_G , r_G (points defining `globalRadialProfile`), `nJoints` (number of joints in the data), b_w (expected bead width)

Output: `jointCoordinates`

Compute length of reference line l using b_w .

Initialise an array $d_{\max}$ to negative infinity to store the maximum deviation for each point along profile.

for point of index j in `globalRadialProfile` **do**

 Find the point of index j_n approximately l away from j .

 Fit a straight line between j and j_n .

 Interpolate the radial coordinates along the line at vertical coordinates defined by z_G .

 Compute the difference between r_G and the interpolated radial values along the line.

 Find the maximum positive signed deviation $d_{\max,j}$ between the line and the global radial profile, and its location $j_{\max}$.

if $d_{\max,j} > d_{\max}(j_{\max})$ **then**

$d_{\max}(j_{\max}) = d_{\max,j}$

end if

end for

Apply a moving maximum filter to $d_{\max}$.

Determine points of local maxima of $d_{\max}$.

Find `jointCoordinates`, the vertical coordinate of the `nJoints` largest local maxima of $d_{\max}$.

5.4.4 Joint segmentation

Can segmentation relies upon correctly segmenting the joints. Two types of joints are present in the WTST dataset, welded joints characterised by the presence of a weld bead and bolted connections between flanges. One method of weld bead segmentation is that of Ye *et al.* (75) where an Expectation Maximisation algorithm was used to iteratively fit a polynomial to the base metal excluding the weld bead. Subtracting the fitted polynomial from the bead profile and applying a threshold height allows for the butt weld bead width and height to be determined. Whilst this method can work well, it is not universally applicable to all WTST scenarios such as flange-to-flange connections where there is no weld bead.

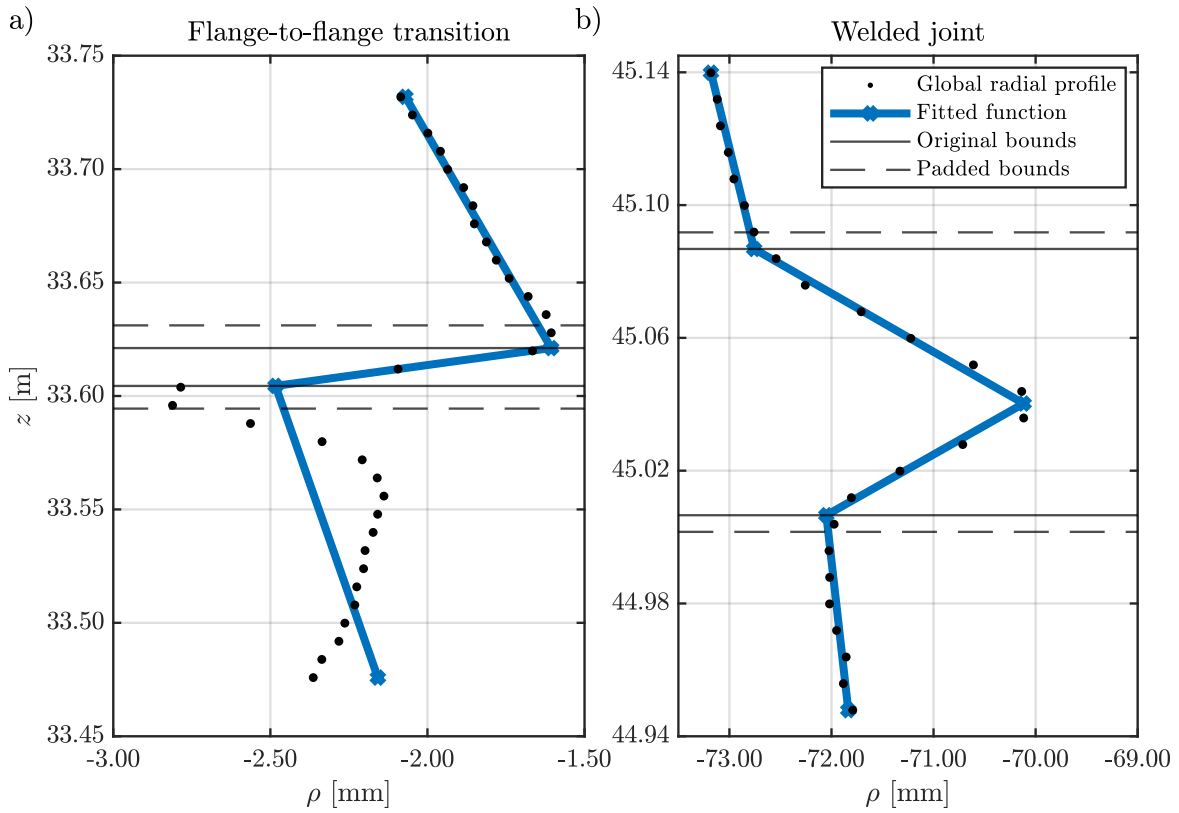


Figure 5: Example of Stage 2 joint segmentation. a) Three-part fit for flange-to-flange transition. b) Four-part fit for welded joint.

An alternate method is proposed in this paper involving fitting piecewise functions to joint regions. Flange transitions and weld beads are modelled using three-part and four-part piecewise linear functions respectively. Given the circumferential variability exhibited at joints due to the welding conditions, circumferentially varying joint eccentricity, varying levels of scan quality etc., the local shape of the point cloud at a particular joint does not have a clearly defined shape which hinders curve fitting and makes isolating the joint region locally in an automatable manner difficult. Instead, the global radial profile offers a good starting point for joint segmentation as weld beads and flange-to-flange transitions are typically easier to identify as seen in Figure 4. Fitting to a single

profile for the entire joint then allows for a single coordinate for the top and bottom of the joint region to be extracted which can then be used to remove the relevant regions at the top and bottom of each can. An example three-part function fitted to a flange-to-flange connection transition and a four-part function fitted to a weld bead is presented in Figure 5a) and b) respectively. Classification of joints as either ‘welded’ or ‘flange-to-flange’ to determine which fit to apply is possible using the nominal information or by inspection of the data. Piecewise functions with both three and four line segments were fitted to the data via a genetic algorithm using the ‘ga’ command of MATLAB. Genetic algorithms were found to yield better fitting solutions when tested against alternative solvers such as a nonlinear least squares solver. The objective was to maximise the coefficient of determination, R^2 . Constraints were applied to enforce that the line did not overlap with itself and that each fitted vertical coordinate corresponded to a single radial coordinate.

To ensure the entire joint region is removed, the identified bounds of the joint can be extended as illustrated on Figure 5. This is particularly applicable to the illustrated flange-to-flange transition where there are some local deviations of the global radial profile from the fitted line segments. Flanges are unlikely to contain sharp structurally relevant features such as dimples and instead typically have more long wave effects such as ovalisation due to their high thickness which will still be present after a more conservative joint segmentation. For the example WTST dataset, flange-to-flange and welded joint regions are extended by 10 mm and 5 mm respectively. Situations can arise where the joints may not be successfully segmented, for example at welded connections between plates with significant thickness differences where the weld bead can be difficult to capture using a four-part piecewise function. The algorithm can also struggle when the features are not clearly defined in the global radial profile such as when there is increased noise in the point cloud towards the top of a WTST. Because a genetic algorithm is used for fitting, the outcome is stochastic in principle. For reproducibility, the released MATLAB workflow fixes the random-number seed, so repeated executions with identical inputs and the same seed reproduce the same fit. Users wishing to explore alternative fits may change the seed. To test the potential variability in outcome arising from the use of a genetic algorithm, each joint region was fitted 100 times using independent random-number seeds, and the standard deviation in each joint coordinate was computed. This resulted in a maximum standard deviation of just 1.7 mm. The joint was also successfully isolated 100% of the time indicating the low variability of the proposed methodology.

5.5 Stage 3: Systematic and random outlier removal

Systematic and random outliers exist in all scanned datasets. Random outliers are points far from the point cloud and do not relate to physical features captured by the scanner and can arise from noise at the edge of a scan where the angle of incidence is largest (68). Systematic outliers relate to legitimate physical features picked up by the scanner which however do not relate to the true shell

outer structural surface such as circumferential and meridional weld beads. Systematic and random outliers must be removed from the dataset as they do not correspond to the true shell outer surface and would otherwise contaminate the Stage 4 reconstructed surface presented in Section 5.6 reducing the reliability of any subsequent finite element modelling or fabrication tolerance assessments. An example of the random isolated outliers and the meridional weld bead is presented in Figure 6 in cylindrical coordinates with the vertical coordinate going into the page. Circumferential weld beads have by now been removed in Stage 2 during can segmentation, however meridional weld beads may still be present in the data. Additional features which are not on the shell surface such as the surrounding environment should have also been removed from the point cloud by the scanning contractor in Stage 0.

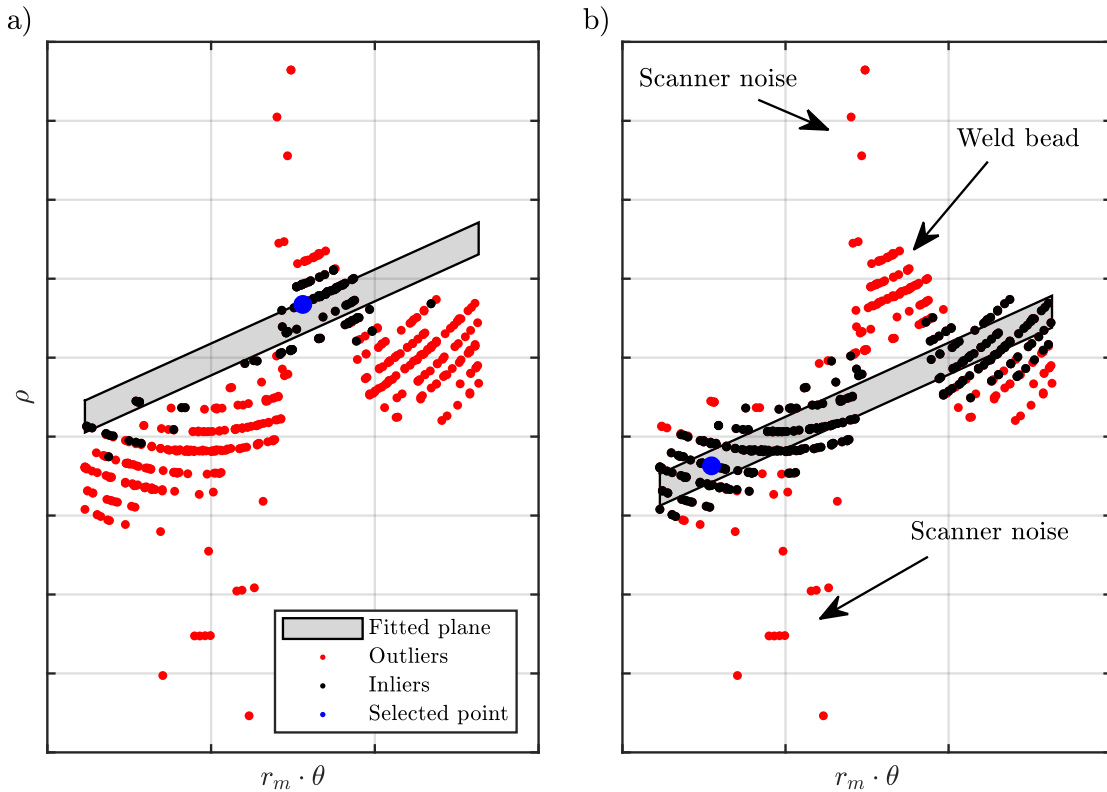


Figure 6: Example of Stage 3 outlier removal with plane fitting.
a) Original fitted plane. b) Corrected plane.

The methodology of Ning *et al.* (2018) (76) for correcting non-isolated outliers is adapted in this paper. Their original method involves finding the k nearest neighbours (denoted q_j) for every point p_i in the point cloud and fitting a local plane through them using principal component analysis (PCA). The distance of q_j from the plane was computed and then q_j was projected onto the local plane to obtain q_j' which became the new corrected coordinate of q_j . The described method involved correcting data points, but the authors highlighted that this may deform the geometry depending on the number of nearest neighbours used. However, for high-fidelity modelling of axisymmetric

shells it is undesirable to have an excessive and unrealistic deformation of the point cloud and thus is preferable to remove outliers rather than correct them. Their algorithm also applies to general point clouds of any shape, but the axisymmetric nature of shells such as WTSTs lends to a more tailored implementation of the algorithm, as described below.

Outlier detection is performed here at the level of the segmented can in cylindrical coordinates. The angular cylindrical coordinate θ of each cloud point (in radians) is multiplied by some known radius r_m (e.g. mean nominal or best-fit can radius) to convert it to a distance to preserve the approximate spacing between points (Figure 6). Contrary to the method proposed by Ning *et al.* (76), instead of finding the nearest neighbours to every point in the cloud, a computationally expensive task, only points within a sliding window of width w_w and height w_h are considered. An arbitrary point p_i of this window is selected and all points in the window are then treated as nearest neighbours. A plane passing through p_i representing the cloud locally is found using PCA and then used to compute the projected distance of points from the plane. Points with a projected distance larger than a cutoff distance from the fitted plane are considered outliers.

Algorithm 4: Systematic and random outlier removal

Input: ρ , θ , z (points in cylindrical coordinates), cutoff (cutoff distance), w_w (window width), w_h (window height) and rmean (average radius)

Output: outlierFreePoints in cylindrical coordinates

Scale circumferential coordinate by rmean.

Extend points circumferentially.

Compute circumferential and vertical coordinates used to define sliding windows.

Initialise an array distpq to store the maximum projected distance of every point to its plane.

for window in slidingWindows **do**

 Identify points which belong to the window.

 Perform PCA to identify the surface normal.

 Select any point (e.g. the first point) through which the plane passes.

 Project points onto the plane and compute the distance to the plane.

 Find the point with the median projected signed distance to the plane and redefine the plane to pass through this point.

 Compute the new projected distance of points to the new plane.

 For every point in the window, update distpq with the new signed projected distance if its absolute magnitude is larger than that of the value currently stored.

end for

Restore circumferential coordinate to its value prior to scaling.

Remove points added to data when extending circumferentially.

Identify outlierFreePoints for which $|\text{distpq}| \leq \text{cutoff}$.

Figure 6a) illustrates the points found in a typical sliding window at a meridional weld bead in the $r_m\text{-}\theta$ to ρ plane (z axis is perpendicular). Marked on the figure are the fitted plane, the initially considered point p_i , outliers and inliers. Figure 6a) illustrates a case where a poor choice of p_i leads to incorrect outlier identification, while Figure 6b) illustrates how this is subsequently corrected (and a better identification of both systematic and random outliers is obtained) when the plane is shifted to the middle of the cloud using the median projected signed distance from the original ‘first estimate’ plane. Points further away than a cutoff distance from this middle plane are considered as candidate outliers, and then removed from the dataset once all possible sliding window positions have been considered. The implementation accepts a scalar cutoff distance, taken as 1 mm for the example WTST. Alternatively, a percentile-derived scalar cutoff may be calculated from a preliminary analysis of the projected distances and supplied to the algorithm. The procedure is summarised in Algorithm 4.

The window is shifted either circumferentially or meridionally by a third of the relevant window dimension to ensure that the results are not dependent on the starting bin location. The size of the sliding window is governed by the size of the features to be retained (e.g., local waves) and to be removed (e.g., weld beads) in the data. A window width $w_w = w_h$ of three times the expected bead width b_w is proposed for WTSTs. Underestimating b_w (and thus the window size) will affect the removal of weld material and may cause issues with fitting a plane during PCA if too few points are in the window and overestimating b_w can lead to overly aggressive outlier removal. The bead width b_w can be estimated based on prior knowledge about typical values for the structure, an on-site physical measurement or by inspection of the point cloud. The data on either side of $\theta = 0$ or 2π must be wrapped around to ensure periodic continuity as this was found to improve the performance of the algorithm at those locations. Based on the above, approximately 15% of the data points were filtered out from the dataset used for development, a portion which depends on the local dispersion within any given point cloud and will thus likely vary for other users.

5.6 Stage 4: Gridded outer shell surface reconstruction of each can

This stage creates a gridded outer structural shell surface representation which can be sampled at arbitrary locations to construct meshes for finite element analysis (which additionally may require further adjustment to obtain the midsurface) or imperfection tolerance assessment as explained in Section 5.8. Many methods of performing surface reconstruction exist in the literature with a comprehensive review of the available methods presented in (77). However, only the comparatively simple method employed by Castro *et al.* (19) and later adapted and called the ‘scatter points’ method by Wang *et al.* (17) appears to be specialised to cylindrical/conical geometries. The method of Castro *et al.* (19) has also been applied by Li *et al.* (18) and can be summarised as follows. First,

a nominal gridded geometry is defined and the k nearest neighbours and corresponding distances d_m from the point cloud is computed for each nominal point. Equation (2) is used to compute the new corrected radial coordinate r_{cor} of each nominal point by averaging the k nearest neighbours with an inverse distance weighting.

$$r_{cor} = \frac{\sum_m^k r_m / (d_m)^P}{\sum_m^k 1 / (d_m)^P} \quad (2)$$

The radial value of nearest neighbours is denoted r_m , P which controls the extent to which distant points affect the weighting was set to 4, and $k = 5$ was adopted by Castro *et al.* (19). Wang *et al.* (17) instead opted to use all nearest neighbours within a certain search radius of a nominal point. A variant of (17) was favoured over other methods in (77) in this protocol given its specialisation to cylindrical/conical geometries. The procedure is presented in Algorithm 5.

In the adopted surface reconstruction algorithm, each segmented can is treated independently as the weld material obscures the true surface and the presence of joint eccentricity means a smooth transition between cans does not exist. As with the scatter points method, first a nominal gridded can is generated at a nominal meridional and circumferential grid spacing g_s obtained as the ‘characteristic data point spacing’ for the can and computed as the 95th percentile of the nearest-neighbour distances for a representative sub-sample of points. The initial nominal gridded geometry is defined by best fitting a truncated cone. Unlike in Stage 1 (Section 5.3), only the top r_t and bottom r_b radius and horizontal x_{bc} and y_{bc} coordinates are solved for as the can is assumed to be correctly orientated. The initial nominal gridded geometry is aligned with the point cloud such that its centre and the centre of the best fitting truncated cone coincide.

For each point on the initial nominal gridded representation, the scatter points method is applied within a search radius. Testing search radii from $4g_s$ to $32g_s$ revealed a relationship with the detail captured in the reconstructed surface. A small search radius can lead to overfitting to the noise in the point cloud whereas a large search radius leads to sharp (and thus more likely to be structurally critical) features of the geometry not being captured. It was found that a small search radius of $4g_s$ with $P = 1$ captured all structurally-relevant features. The reconstructed surface is then smoothed using a 2D Gaussian filter by ‘imgaussfilt’ of MATLAB to handle any spurious gridded points and overfitting to noise, with a standard deviation of 10 and a filter size $f_s = \text{floor}(\lambda / (5g_s))$, though not less than 3, where $\lambda \approx 2.44\sqrt{(rt)}$ is the linear meridional bending half-wavelength. Circumferential periodicity is enforced during filtering to avoid an artificial discontinuity at $\theta = 0, 2\pi$ while replicated values are used at the meridional boundaries.

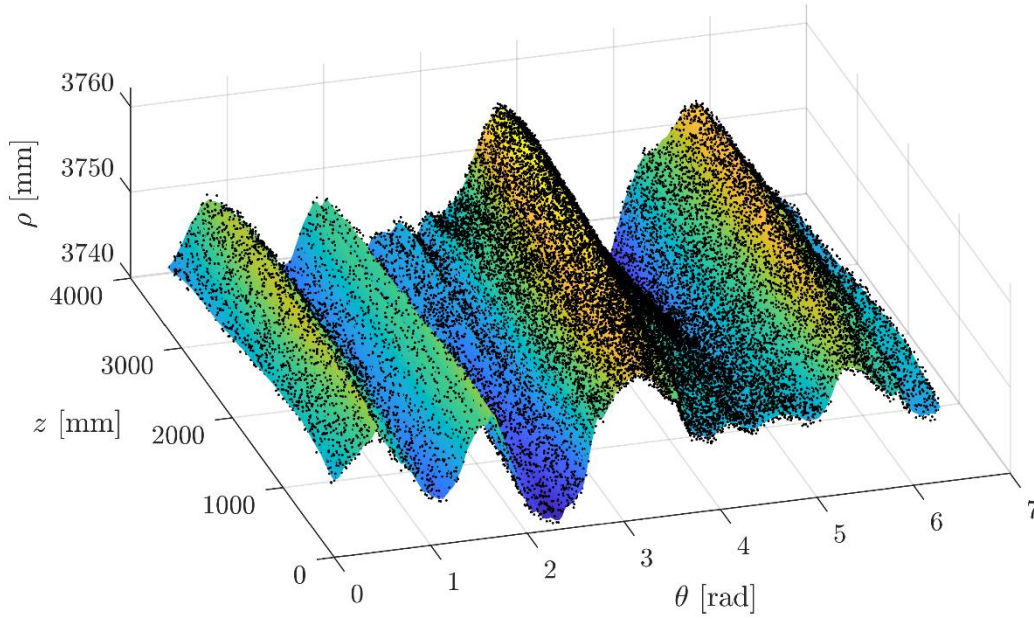


Figure 7: Example of a Stage 4 reconstructed outer surface.

Algorithm 5: Can gridded outer shell surface reconstruction

Input: ρ , θ , z (outlier free points in cylindrical coordinates), topCan and botCan (vertical coordinate of top and bottom of can), nominal can properties, p , searchRadiusGridSpacingSF and smoothingFilterStd

Output: R_g , θ_g , Z_g (reconstructed outer surface grid defined in cylindrical coordinates)

Compute Cartesian coordinates x , y of all points.

Select nRandom points of the can to compute nearest neighbours for.

for point in nRandom points **do**

 | Compute nearest neighbour distance from point to full can dataset and append to minDist.

end for

Set gridSpacing to 95th percentile of minDist.

Solve for the top radius r_t , bottom radius r_b , and two horizontal centroid coordinates x_{bc} and y_{bc} of the best-fitting truncated cone.

Define vertical coordinate z_{ng} for the nominal grid.

Estimate the variation of the radius r_h with height with a linear function using r_t and r_b .

Define circumferential coordinate θ_{ng} for the nominal grid.

Create θ_g , Z_g which are the meshgrid representations of θ_{ng} and z_{ng} respectively.

Create initial nominal X_g and Y_g from θ_g and r_h and shift to best fit centre of point cloud.

Compute searchRadius using gridSpacing and searchRadiusGridSpacingSF.

for circumferential coordinate in θ_{ng} **do**

 | **for** vertical coordinate in z_{ng} **do**

 | Reset searchRadius to initial value.

 | **while** true **do**

 | Find nearest neighbours to chosen nominal point within the defined searchRadius.

```

    if no nearest neighbours are found then
        Double search radius.
    else
        Apply Equation (2) and compute updated grid radial value and store in  $R_g$ .
        break
    end if
end while
end for
end for

```

Compute the Gaussian filter size from the nominal can properties and characteristic grid spacing, enforce a minimum size of 3, and round an even value upwards to the next odd integer.

Apply Gaussian filter with standard deviation `smoothingFilterStd` to R_g for smoothing.

The half-wavelength λ is a result from linear shell bending theory and relates closely to the ‘boundary layer’ of compatibility bending that occurs near can boundaries which attract local bending-related plastification. It also relates closely to the theoretically smallest linear elastic buckles that can occur in a cylindrical shell (which is always a meridional compression buckle). Consequently, λ captures the appropriate geometric scale relative to which various resolutions can be determined, including the present filter size, but also scanning resolution (66) and mesh design (78), particularly if the end use of the point cloud is a high-quality GMNIA (geometrically and materially nonlinear finite element models with explicit inclusion of imperfections). Small gaps in the point cloud such as from the removal of meridional welds in Stage 3 are automatically filled in by this method. The adopted methodology based on (17,19) is presented in Algorithm 5. The key hyperparameters relating to the variable P of Eq. (2), the scale factor applied to grid spacing to inform the search radius and standard deviation for the smoothing filter are inputs to the algorithm and are denoted `p`, `searchRadiusGridSpacingSF` and `smoothingFilterStd` respectively. An example of the reconstructed surface of a single can is presented in Figure 7 (where the grid lines of the mesh have been disabled to ensure the point cloud is visible). The presented method assumes the axisymmetric shell is geometrically similar to its nominal geometry with only small imperfections. Shells which are severely damaged such that a particular circumferential and vertical coordinate no longer relate to a single unique radial point due to the shell being folded over violate the assumptions of the algorithm and will not be correctly reconstructed.

5.7 Stage 5: Surface inpainting

The raw point cloud is very likely to have missing regions arising from either gaps in the data such as from obstacles blocking the line of sight of the laser and locations which are difficult to scan such as the upper regions of tall towers as seen in Figure 1e). In regions of partial laser scan coverage where the shell can credibly be assumed to be tolerance-compliant, machine learning

techniques may be employed to perform point cloud inpainting and provide a reasonable estimation of the geometry allowing more of the shell to be modelled. Inpainting methods for large but benign regions with missing coverage are still under development and are beyond the scope of this paper, but an example of a technique that may be employed is presented in (79). If the user is applying point cloud inpainting, this stage would occur before surface reconstruction. Instead, if the gridded surface is being inpainted, as in this paper, then this stage occurs after surface reconstruction.

The processing of scan data to remove circumferential (Stage 2) and meridional weld beads (Stage 3) also creates regions of missing coverage, and the true shell surface underneath the weld beads is anyway unknowable with laser scanning. Yet data in this region can have significant implications for both the finite element buckling assessments and fabrication tolerances, as joints are often critical locations for buckling or tolerance excesses. Plate eccentricities at meridional weld beads occurring in the middle of a can are likely to be small and can be automatically filled in during surface reconstruction (Stage 4). Teng *et al.* (30) removed circumferential weld beads from silo measurements using a cubic interpolation scheme, however such an approach masks any plate eccentricity as it results in a single radial coordinate at joints when there are in fact two (one for each can), and this approach was thus not adopted here. Peng *et al.* (21) instead fitted a Fourier series to fill gaps left by weld bead removal but speculated that it may have led to discrepancies when trying to recreate laboratory tests of cylinders under bending using finite element models with the real geometry.

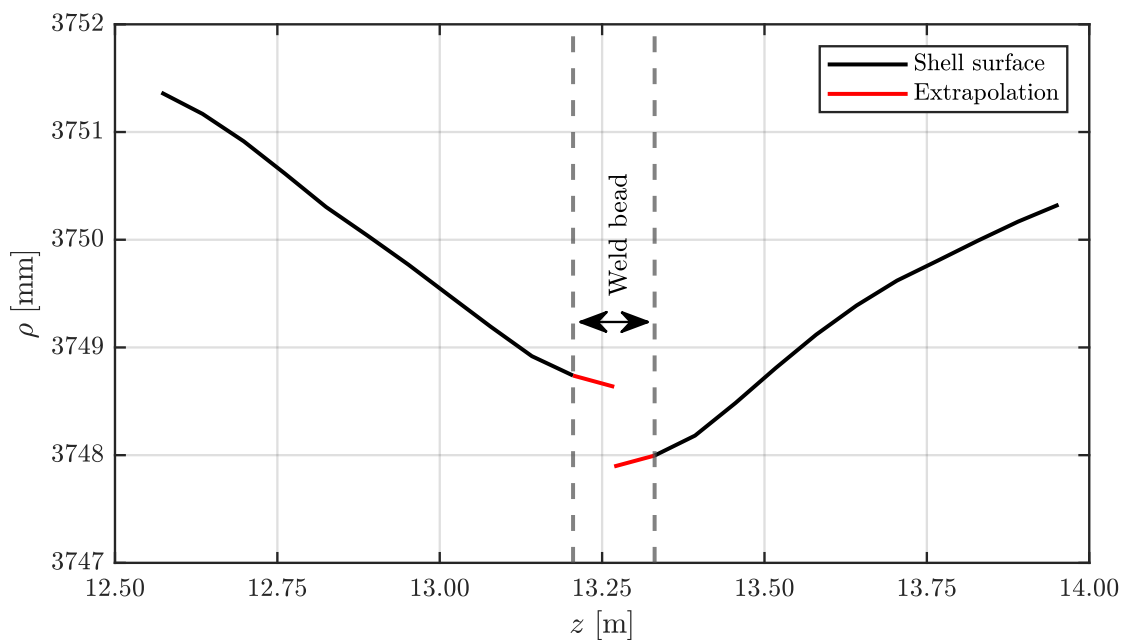


Figure 8: Example of a Stage 5 inpainted joint profile.

For now, a simple algorithm involving linear extrapolation is proposed and is presented in Algorithm 6. Since there is a discontinuity in the shell outer surface at the joint, the plates on either side of the joint are treated independently. The last two gridded points of a chosen meridian in the vicinity of the joint for a can are used to guide the linear extrapolation. The gradient of the line between the two points is maintained, and the shell is extended from the end of the can to the centre of the joint as this is where the plate is assumed to terminate. For simplicity each meridian is treated independently despite the shape of adjacent meridians being linked. An example of the extrapolation scheme is illustrated on a single meridian in the vicinity of a weld bead in Figure 8. Alternate extrapolation schemes including fitting a second order polynomial were tested, but linear extrapolation appeared to work best based on a visual examination of the inpainted surface. Algorithm 6 is applied to all circumferential joints for every meridian of each can.

5.8 Stage 6: Projection to arbitrary meshes

The inpainted reconstructed representation of the shell outer surface available at this stage for each can is stored on a regular rectangular grid with a fine grid spacing resolution g_s given by the ‘characteristic data point spacing’ (Stage 4), this being the finest resolution justifiable by the scanning resolution. A clean outer surface is thus now available for further analysis, which may be either a statistical exploration of the surface geometric imperfections or structural finite element analysis. In either case, the analyst may require a different ‘mesh’ than that offered by this representation, and an additional step to facilitate this is required where the grid representation must be projected onto a different (and not necessarily uniformly-spaced) desired mesh.

For computational GMNIA shell buckling calculations, the finite element meshing requirements of Sadowski (78,80) are used in this protocol to generate a mesh compliant with buckling modes predicted by the ‘Koiter ellipse’, which are the smallest theoretically possible buckling modes on a given shell geometry. The meridional mesh resolution is free to vary between cans, but the same circumferential resolution is applied to each can (chosen as the finest one across all cans). A consistent circumferential spacing can be useful for buckling-relevant fabrication tolerance assessment (presented shortly) as it allows for easy identification of and access to individual meridians (as they share the same circumferential coordinate and relate to a single column of a matrix) and can be helpful for setting up finite element models. A uniform element spacing is not necessary and the analyst is free to specify any mesh bias. The projection operation is generic and may be applied to arbitrary target nodal coordinates; the released MATLAB implementation constructs a Koiter-ellipse-compliant target mesh internally as a good-practice example. The released implementation samples the regular Stage 5 surface onto a target mesh in cylindrical coordinates using linear interpolation with the ‘griddedInterpolant’ command of MATLAB. An

example of this procedure is presented in Figure 9a) and b) which relate to a uniform mesh and a mesh with meridional variation being sampled from a Stage 5 surface. The sampled meshes can be used in conjunction with an input file generator to make finite element models for simulation in third-party software. For finite element models it may be advantageous to store the sampled mesh of each can independently as different cans may have different properties such as plate thickness, while for fabrication tolerance assessment it may be more advantageous to store the coordinates of all cans in one large continuous matrix. Provided plate thicknesses are close to nominal values, the shell midsurface can be obtained if desired.

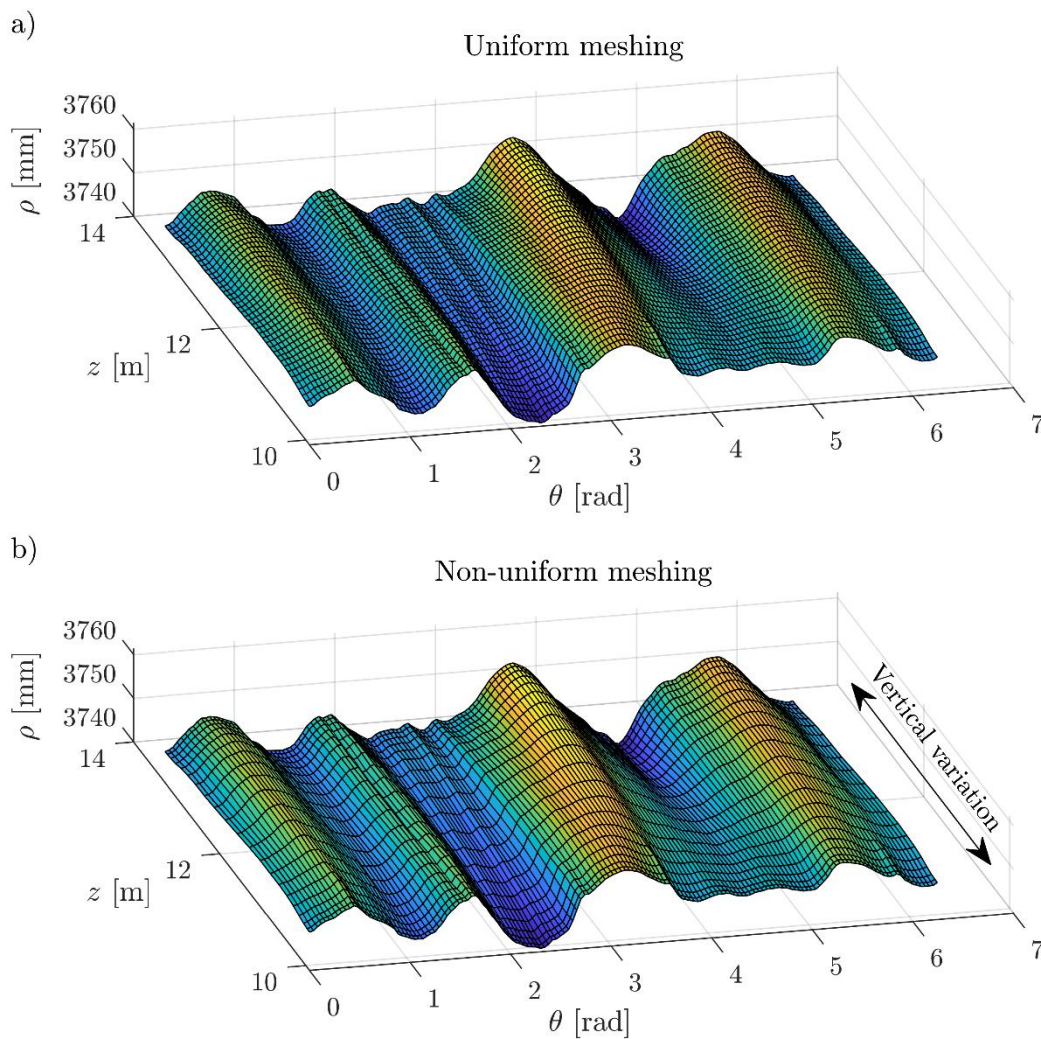


Figure 9: Examples of Stage 6 finite element meshes sampled from reconstructed surface. a) Uniform mesh. b) Non-uniform mesh.

Algorithm 6: Joint inpainting

Input: rv , zv (meridian radial and vertical coordinates), $top2JointCentre$ (distance from top of can to centre of joint), $bot2JointCentre$ (distance from bottom of can to centre of joint)

Output: rv , zv (meridian radial and vertical coordinates)

Add $top2JointCentre$ to the top vertical coordinate to yield $zvTop$ at the joint centre.

Extrapolate the radial coordinate $rvTop$ to the top of the meridian, $zvTop$.

Take $bot2JointCentre$ from the bottom vertical coordinate to yield $zvBot$ at the joint centre.

Extrapolate the radial coordinate $rvBot$ to the bottom of the meridian, $zvBot$.

Create new arrays $rv = [rvBot \text{ } rv \text{ } rvTop]$, $zv = [zvBot \text{ } zv \text{ } zvTop]$.

Correct vertical coordinates zv such that they start at zero.

973

974 6. Buckling-relevant fabrication tolerance assessment

975 6.1 Introduction

976 The recently redeveloped state-of-the-art Eurocode EN 1993-1-6 (2025) (65) on the strength and
977 stability of metal shells requires each constructed shell structure to be assessed for four buckling-
978 relevant fabrication tolerances (dimple, out-of-roundness, unintended eccentricity and interface
979 flatness). Earlier versions of the standard had envisioned that this would be primarily done
980 manually using physical gauges (sticks), but in view of the laborious and error-prone nature of
981 manual assessments (and the difficulty and danger in accessing remote parts of the structure), the
982 2025 standard recognises that assessments will increasingly be done computationally on point
983 clouds obtained from full-scale laser scanning (or other reality capture). To this end, this paper
984 presents novel algorithms to automate the computational buckling-relevant tolerance assessment
985 together with their MATLAB implementation. In addition to finite element mesh generation for
986 computational shell buckling analysis with GMNIA, this fabrication quality assessment thus
987 represents a second major use case for the geometric digital twinning protocol just presented, and
988 the algorithms presented here for this purpose are believed to be the first of their kind.

989 The dimple, out-of-roundness and unintended eccentricity tolerances each relate to predominantly
990 radial imperfections in the shell wall and are assessable with an outer surface laser scan of a
991 fabricated shell, while the interface flatness tolerance relates to a meridional imperfection at
992 boundaries typically found at the base of a shell structure. For WTSTs, this flatness relates to
993 imperfections in the flange mating surface and should be assessed by laser scanning the appropriate
994 flange mating surfaces before assembly. Since the WTST data present only relates to the radial
995 imperfections in the shell, an algorithm for this tolerance is not presented in this paper (though one
996 can readily be derived based on the concepts presented here). The algorithms presented here are

applied to individual meridional or circumferential generators of the gridded representation of the gDT from Stage 6 of the protocol. The application of these algorithms does not strictly require an initial point cloud representation of a shell to be passed through the protocol into a gDT provided the relevant information is supplied. To the best of the Authors' knowledge only Kathirkamanathan, Sadowski and Seidel (81) have hitherto formally presented any systematic method for assessing the meridional dimple tolerance, to be superseded by the algorithms presented here which are more accurate and efficient.

6.2 Meridional dimple tolerance assessment

6.2.1 Introduction

The meridional dimple tolerance measures the deviation of a meridian from its hypothetical nominal geometry. This tolerance can be measured on site using a physical gauge which is placed against the shell wall, the gauge creating a local reference axis relative to which deviations can be assessed. The perpendicular distance between the gauge and the shell is measured and normalised by the gauge length to determine the meridional dimple tolerance. EN 1993-1-6 (2025) (65) provides provisions for determining the length of the gauge as a function of the local destabilising membrane stress state and the local radius to thickness ratio. The achieved tolerance values can be compared against tabulated values to describe the fabrication quality as one of three Fabrication Tolerance Quality Classes (FTQC): A ('excellent'), B ('high') or C ('normal').

Algorithm 7: Gauge-Snapping Algorithm

Input: rv , zv (meridian coordinates) or xv , yv , $thetv$ (circular arc coordinates), j (gauge start index), $rnom$ and $tnom$ (nominal radius and thickness at j)

Output: Dimple tolerance

Call `defineGauge` (Function 7.1 (meridional gauge) or 7.5 (circumferential gauge))

while true do

 Call `computeShellGaugeCorrespondences` (Function 7.2)

 Call `applyGaugeRotations` (Function 7.3)

 For the circumferential gauge, call `selfIntersectionCheck`

if the gauge does not intersect the shell wall between its contact points **then**

 Call `measureTolerance` (Function 7.4 (meridional gauge) or 7.7 (circumferential gauge))

end if

if end of shell or gauge reached **then**

 | break

end if

end while

6.2.2 The Meridional Gauge-Snapping Algorithm

Measuring the meridional dimple tolerance computationally is challenging as the process needs to mirror the measurement that would occur physically on site. Any algorithm must consider collisions of the simulated gauge with the shell wall and ensure that the simulated gauge does not pierce the shell at any point. A general Gauge-Snapping Algorithm is presented in Algorithm 7 which may be used to assess both the meridional and circumferential dimple tolerances. The key steps of the algorithm have been abstracted into functions to aid readability and highlight the common framework adopted for the meridional and circumferential dimple tolerances (presented in Section 6.3). EN 1993-1-6 (2025) (65) prescribes a straight gauge to assess the meridional dimple tolerance which is employed here, although the algorithm is structured such that the user can define a gauge shape which is more representative of the nominal meridional geometry i.e. a curved gauge for spherical shells, and with the relevant modifications assess the tolerance for other geometries. The procedure is illustrated in Fig. 10 on a single gauge start position for a single meridian, though in practice it must be applied at all possible start positions for all meridians to compute the tolerances for the entire shell. A coordinate system with the radial coordinate ρ on the abscissa and the vertical coordinate z on the ordinate is adopted. The inputs are the gauge start index j along the target meridian, radial r_v and meridional z_v coordinates of the meridian each 1D arrays of equal size, nominal radius r_{nom} and thickness t_{nom} at j . The output is the meridional dimple tolerance U_{0x} measured for all positions of the gauge with start index j .

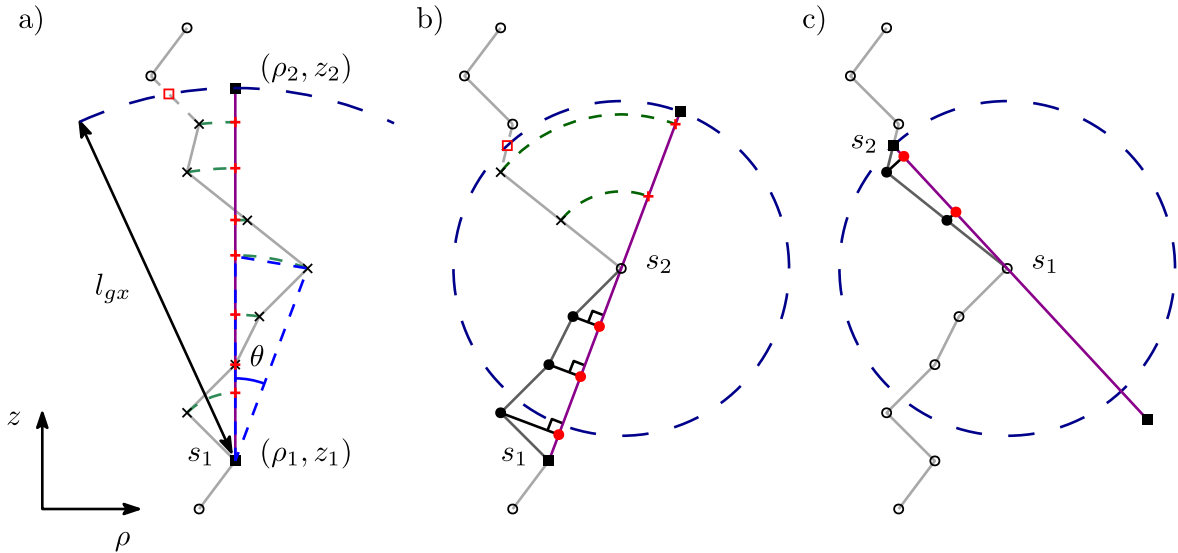


Figure 10: Meridional Gauge-Snapping Algorithm. a) Initial gauge position.

b) First snap. c) Second snap.

A point is first chosen along the shell profile (grey line) which represents the start of the gauge (ρ_1, z_1) and is marked with a filled black square on Figure 10a), with points along the meridian marked with empty black circles. The gauge (magenta line) is initially assumed to be orientated vertically and its Eurocode-compliant length $l_{gx} = 4\sqrt{rt}$ (designed to capture short-wave local

features sympathetic to meridional buckling modes) is computed using the nominal radius r and thickness t at the start of the gauge. A gauge of length l_{gw} is to be used instead under conditions outlined in EN 1993-1-6 (2025) (65), though this alternative is not currently implemented in the MATLAB code. The computed coordinates of the end of the gauge (ρ_2, z_2) is marked with a filled black square on Figure 10a), and the start of the gauge is the first snapping point denoted s_1 . This described functionality is performed by Function 7.1. In subsequent steps, the gauge will rotate about s_1 until it contacts the shell wall again at a second point described as the second snapping point s_2 . The process of the gauge jumping between possible configurations as it rotates along the shell wall is here described as ‘snapping’, and each point of intersection has been described as a ‘snapping point’ in this algorithm.

The distance between s_1 and the end of the gauge is computed and points within this distance from s_1 are considered as feasible locations for the gauge to contact the shell after rotation and are marked with black crosses on Figure 10a). The corresponding points on the gauge which are the same distance from s_1 as the points on the shell are marked with red pluses. Line segments along the shell wall where the end of the gauge could contact are also considered (marked as the dashed grey line). The point where the gauge could contact along the line segment is marked with a red box on Figure 10a). The action of computing potential correspondence between the shell and the gauge for rotation is performed by Function 7.2. Next the angle of rotation for each possible point is computed and an example is marked with a blue triangle on Figure 10a). The maximum and minimum of the considered signed angles of rotation correspond to rotated gauge positions with a single secondary contact with the inside and outside of the shell wall respectively. The desired rotation of the gauge about s_1 resulting in a single secondary contact with the shell wall at s_2 is determined and the gauge rotated to its new position shown in Figure 10b). A check is performed to see if s_2 is at the end of the gauge or the end of the shell in which case the gauge cannot be further rotated. An example of such a scenario is presented in Figure 10c) where the end of the gauge coincides with s_2 and meets the shell wall. If further rotation is possible, s_1 is set to s_2 . The relevant pseudocode is presented in Function 7.3.

$$\begin{pmatrix} \rho_s \\ z_s \end{pmatrix} = \begin{pmatrix} \rho_1 \\ z_1 \end{pmatrix} + \xi \begin{pmatrix} c \\ d \end{pmatrix} + \delta_{0x} \begin{pmatrix} d \\ -c \end{pmatrix} \quad (3)$$

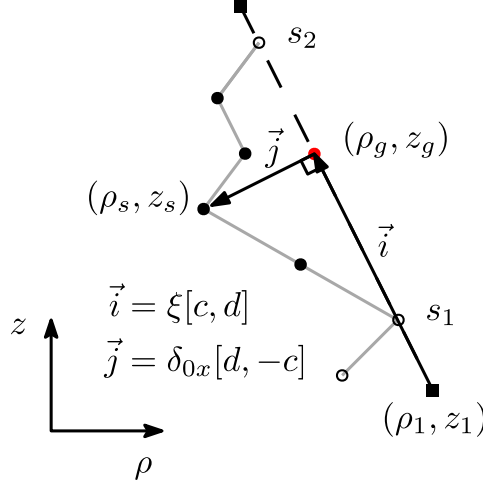


Figure 11: Measurement of a meridional dimple tolerance.

The perpendicular distance between the shell and the gauge is measured by solving the simultaneous equations presented in Eq. (3) for δ_{0x} which is depicted in Figure 11. The normalised vectors $[c \ d]$ and $[d \ -c]$ are respectively parallel to and perpendicular to the gauge, (ρ_s, z_s) is a point on the shell while (ρ_g, z_g) is the point on the gauge where the line perpendicular to the gauge passing through (ρ_s, z_s) originates from and is marked with a filled red circle. The distance between (ρ_1, z_1) and (ρ_g, z_g) is denoted ζ while δ_{0x} is the perpendicular distance between the shell and the gauge (i.e. between (ρ_g, z_g) and (ρ_s, z_s)). Correspondences between points on the shell and the gauge for tolerance measurement are marked on Figure 10b) with lines between the two. The distances between relevant points on the shell wall and the gauge δ_{0x} are determined and normalised by the gauge length l_{gx} to yield tolerance measurements as $U_{0x} = \delta_{0x} / l_{gx}$. The points (ρ_g, z_g) are checked to see whether they truly lie on the gauge between s_1 and s_2 using ζ and corresponding U_{0x} values are removed otherwise. The pseudocode for tolerance measurement is presented in Function 7.4.

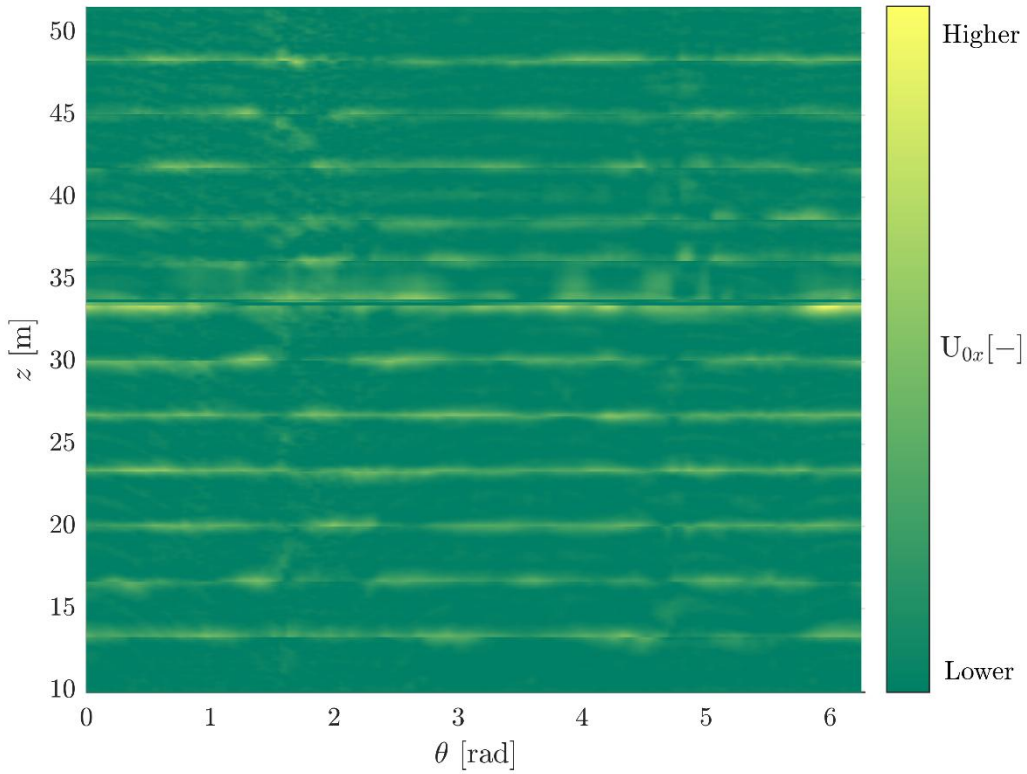


Figure 12: Meridional dimple tolerance evaluated over example WTST data.

After the tolerance is measured, provided s_2 is not at the end of the gauge or the end of the shell, the gauge is rotated about its updated s_1 and the tolerance measurements extracted again as shown in Figure 10c). The gauge is rotated along the shell wall until either the end of the gauge contacts with the shell wall or the top of the shell is reached. The start point of the gauge is then moved along the shell wall, and the procedure repeated. For a single point on the shell wall, the maximum of the tolerances measured for different gauge positions at that point is taken as the dimple tolerance for that point. An example of the Gauge-Snapping Algorithm for the meridional dimple tolerance being applied to every point on the Stage 6 meshed representation of the example WTST dataset is presented as a ‘heat map’ in Figure 12 assuming a gauge of length l_{gx} . The heat map shows increased tolerance measurements at the circumferentially welded joints, as expected. In situations where there are variations in the nominal radius and thickness with height such as in shallow conical shells, users must be careful with the choice of l_{gx} . Using the nominal properties at the bottom of the gauge lead to a larger l_{gx} and thus a potentially smaller and less conservative tolerance measurement when calculating U_{0x} . In such situations the user could modify the algorithm such that the top of the gauge starts in contact with the shell, the corresponding nominal properties are used to define l_{gx} , and the gauge is rotated until the bottom of the gauge meets the shell.

Function 7.1: defineMeridionalGauge (shellPoints, rnom, tnom, j)

Compute l_{gx} using the nominal properties at j.
 Initialise gaugeVector = $[0 \ 1]^T$ to represent the orientation of a vertical gauge.
 Extract coordinates of gaugeStart from shellPoints.
 Compute coordinates of gaugeEnd.
 Initialise snapOneInd, to j representing the start of the gauge index.
return l_{gx} , gaugeStart, gaugeEnd, snapOneInd

1101

1102 **6.3 Circumferential dimple tolerance assessment**

1103 The circumferential dimple tolerance measures the deviation of the circumference from its
 1104 nominally circular geometry using a curved gauge. The Gauge-Snapping Algorithm presented in
 1105 Algorithm 7 is also used to measure this tolerance, with appropriate modifications for curved
 1106 gauges as detailed here. The circumferential dimple tolerance measurement requires as input the
 1107 coordinates that define the circular arc i.e. Cartesian coordinates x_v and y_v as well as the
 1108 corresponding cylindrical coordinate θ_v denoting the circumferential position in radians. All
 1109 functions have similar pseudocode and inputs to the meridional dimple tolerance formulation and
 1110 thus are not repeated here, with the exception of Function 7.5 which replaces 7.1, Function 7.7
 1111 which replaces 7.4 and an additional Function 7.6 which creates the `gaugePoints` variable used
 1112 by Function 7.2.

1113 EN 1993-1-6 (2025) (65) requires that circumferential dimple tolerances should be assessed against
 1114 two gauge lengths: the short-wave $l_{gx} = 4\sqrt{rt}$ as before (to check for dimples sympathetic to short-
 1115 wave meridional buckling modes) and the much longer $l_{g\theta} = \max(r, 2.3(l_s^2 rt)^{0.25})$ gauge (which
 1116 checks for dimples sympathetic to long-wave torsional or circumferential buckling modes), where
 1117 l_s is the length between shell segment boundaries i.e. flanges. In the released illustrative MATLAB
 1118 implementation, l_s is approximated by the height of the corresponding segmented can. This
 1119 approximation affects only the long-wave circumferential gauge $l_{g\theta}$; for a formal standards-based
 1120 assessment, l_s should instead be supplied from the design information as the applicable boundary-
 1121 to-boundary shell-segment length. The radius of curvature of the gauge and intended shell
 1122 midsurface must also be equal. Additionally, the presented algorithm should not be applied to
 1123 gauges bigger than a semicircle. However, this requirement can easily be satisfied for most full-
 1124 scale shell structures when using the aforementioned gauge lengths. The algorithm must only be
 1125 applied when the circumferential gauge lies on the outside of the shell. Tolerances are calculated
 1126 as $U_{0x} = \delta_{0x} / l_{gx}$ or $U_{0\theta} = \delta_{0\theta} / l_{g\theta}$.

Function 7.2: computeShellGaugeCorrespondences (shellPoints, gaugeStart, gaugeEnd, snapOneInd)

Initialise searchRange, the indexes of shellPoints to check from snapOneInd until the end of the shellPoints.

Initialise snapOne, the coordinates of the point about which the gauge will rotate.

Compute snapOne2EndLength, the distance between snapOne and gaugeEnd.

Find snapTwoPos, the indexes of points of shellPoints with an index greater than or equal to snapOneInd within snapOne2EndLength of snapOne where the gauge could rotate to.

Extract coordinates of points along shell wall, snapTwoPosPoints, with index snapTwoPos.

Compute points, gaugePoints, along the gauge with the same distance from snapOne as snapTwoPosPoints.

Find lineSegments, the line segments along the shell where the end of the gauge could possibly land. These are stored as pairs of indexes.

for lineSegment in lineSegments **do**

 Solve analytically for the intersection between the lineSegment and a circle of radius snapOne2EndLength and origin snapOne and store in array intersects.

end for

Append intersects to array snapTwoPosPoints, and gaugeEnd to gaugePoints for each intersection of intersects found if any.

return lineSegments, intersects, snapOne, gaugePoints, snapTwoPos, snapTwoPosPoints, searchRange

1127

Function 7.3: applyGaugeRotations (gaugePoints, snapTwoPosPoints, snapOne, snapTwoPos, searchRange, intersects, gaugeStart, gaugeEnd, shellPoints, lineSegments, side)

Compute the angle enclosed by vectors from snapOne to gaugePoints and snapOne to snapTwoPosPoints.

Determine the appropriate angle to use based on its sign and whether the inside or outside of the shell is being assessed.

Rotate gaugeStart and gaugeEnd about snapOne.

Determine snapTwo, the second point of contact of the gauge with the shell wall.

Determine tolMeasureRange, the indexes of points where tolerance measurements are feasible.

Determine the boolean endReached whether the end of the shell or gauge has been reached.

if not endReached **then**

 Update snapOneInd with index of snapTwo.

end if

return gaugeStart, gaugeEnd, tolMeasureRange, snapTwo, snapOneInd, endReached

1128

Function 7.4: measureMeridionalTolerance (gaugeStart, gaugeEnd, shellPoints, snapOne, snapTwo, tolMeasureRange, lgx)

Determine the perpendicular distance between the shell and the gauge.

Check points on gauge to see if they truly lie between snapOne and snapTwo and remove incorrect points.

Normalise perpendicular distances by the gauge length to yield the toleranceValues.

return toleranceValues

Function 7.5: defineCircumferentialGauge (shellPoints, rnom, gaugeLength, j)

Extract coordinates of gaugeStart from shellPoints

Initialise snapOneInd, to j representing the start of the gauge index.

Set the gauge length to the prescribed gaugeLength, previously computed as l_{gx} or $l_{g\theta}$ from the applicable nominal properties.

Compute angle swept by the gauge θ_l .

Compute normalised vector from gaugeStart to nominal cross-section centre (0,0)

Compute coordinates of gaugeCentre assuming it lies on vector from gaugeStart to (0,0)

Compute searchLimit, the index of the point π radians away from snapOneInd

return l_{gx} or $l_{g\theta}$, θ_l , gaugeStart, gaugeCentre, snapOneInd, searchLimit

$$\beta = 2 \sin^{-1} \left(\frac{\gamma}{2r} \right) \quad (4)$$

$$\text{gaugePoint} = r \cdot \begin{pmatrix} \cos(\theta + \beta) \\ \sin(\theta + \beta) \end{pmatrix} + \text{gaugeCentre} \quad (5)$$

When computing gaugePoints in Function 7.2, points along the gauge which are the same distance from s_1 as points on the shell are required. The gaugePoints have been computed using Function 7.6 as illustrated on Figure 13a) by applying Eqs (4) and (5). As with the meridional dimple tolerance, s_1 is the first point of contact between the shell and gauge and its coordinates are known. The shell is marked with a dashed grey line and the gauge with a circular arc. An example of a point on the shell for which a corresponding point on the gauge is required is denoted (x_s, y_s) on Figure 13a). The distance γ between (x_s, y_s) and s_1 can be used to calculate the enclosed angle β between s_1 and the gaugePoint using Eq. (4). θ is the angular position of s_1 relative to the gaugeCentre (centre of the circle on which the arc defining the gauge lies) marked on Figure 13a) and r is the nominal shell radius (rnom). Applying Eq. (5) then yields the Cartesian coordinates of the gaugePoint. Alternatively, if the angle relative to the gaugeCentre between two points is already known such as when trying to compute gaugeEnd which is θ_l from gaugeStart, then γ is not required and Eq. (5) can be applied directly (in this case with θ being the angular position of gaugeStart relative to the gaugeCentre and $\beta = \theta_l$). Prior to the algorithm being applied to a cross-section, the geometric least-squares best-fitting circle of the cross-section is computed, and the corresponding lateral translations applied to shift the centre of the cross-section onto the origin which helps correctly compute thetv.

The measurement of the circumferential dimple tolerance (Function 7.7) is presented in Figure 13b) and c). The normalised vector from `gaugeCentre` (marked with a black box filled white) to the points on the shell, `shellPoints` denoted (x_s, y_s) is computed. Scaling the vector by the radius of the gauge `rnom` yields the corresponding coordinates of possible points on the gauge (x_g, y_g) used to evaluate the tolerance. Figure 13c) shows an example of two possible points, the correct point marked with a red circle, and the incorrect point marked with a black circle shaded grey. Points which truly lie on the gauge are identified and the distances to `shellPoints` are computed and then normalised by the `gaugeLength` to yield the `toleranceValues`.

Function 7.6: `circumferentialGaugePoints` (`gaugeCentre`, `beta`, `theta`, `r`)

Compute the angular position corresponding to `gaugePoints` i.e. $(\theta + \beta)$.

Apply Equation (5) to compute `gaugePoints` in Cartesian coordinates.

return `gaugePoints`

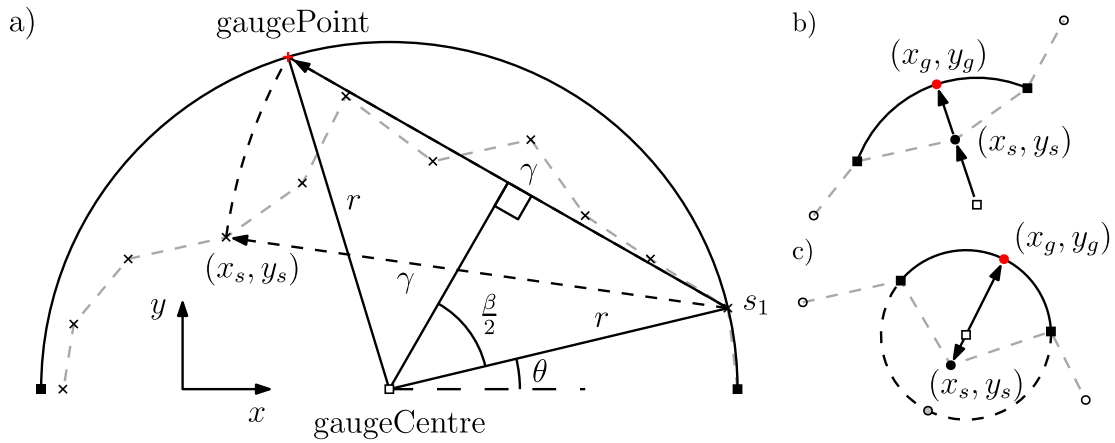


Figure 13: a) Computing potential `gaugePoints` in Function 7.6. b) and c) Computing point on gauge for tolerance measurement.

Function 7.7: `measureCircumferentialTolerance` (`gaugeStart`, `gaugeCentre`, `shellPoints`, `rnom`, θ_l , `gaugeLength`)

Compute normalised vectors `n`, from `gaugeCentre` to `shellPoints`.

Compute possible points on the gauge which are `rnom` from `gaugeCentre` along vectors `n` in both opposing directions.

Determine whether the computed points truly lie on the gauge using `gaugeStart` and θ_l and remove incorrect points.

Determine the distance between `shellPoints` and points on the gauge.

Normalise distances by the `gaugeLength` to yield the `toleranceValues`.

return `toleranceValues`

6.4 Out-of-roundness tolerance assessment

The out-of-roundness tolerance $U_r = (d_{\max} - d_{\min}) / d_{\text{nom}}$ relates to the lack of circularity of a cross-section and can be evaluated by determining the difference in the maximum $d_{\max}$ and minimum $d_{\min}$ diameters at an elevation normalised by the nominal diameter d_{nom} (Figure 14). An algorithm for computing the out-of-roundness is presented in Algorithm 8 and must be applied at every elevation. As stated by EN 1993-1-6 (2025) (65), the diameter of a cross-section from a point is measured as the maximum distance from the point to all other points of the cross-section.

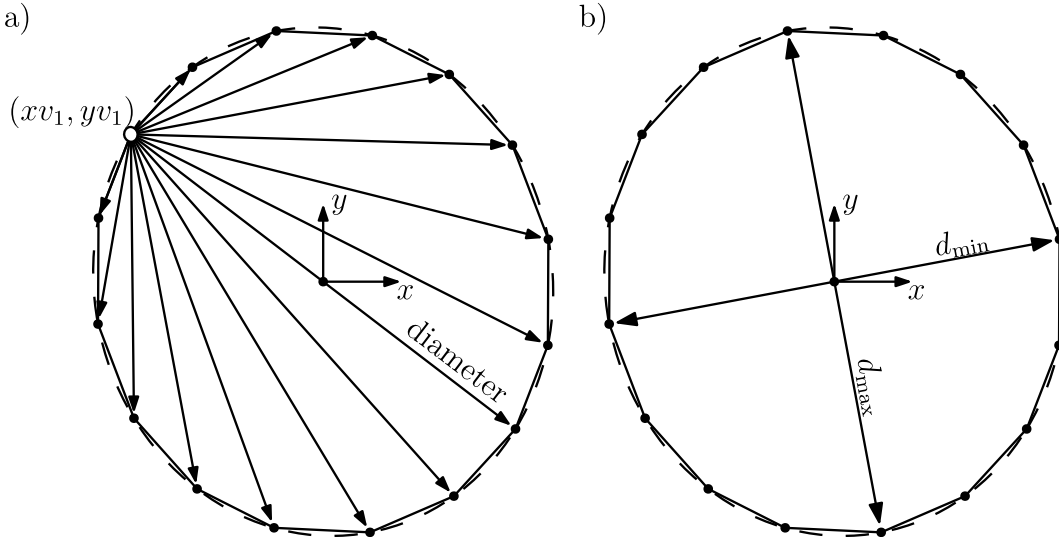


Figure 14: Out-of-roundness measurement. a) Diameter measurement. b) Diameters for tolerance assessment.

Algorithm 8: Out-of-roundness tolerance measurement

Input: xv, yv (Cartesian coordinates of circular arc) and d_{nom} (nominal diameter)

Output: Out-of-roundness tolerance U_r

for point in (xv, yv) **do**

 Compute distance between point and every other point in (xv, yv) .

 Determine maximum distance and store in a 1D array D . This corresponds to the diameter measured from the point.

end for

Extract the smallest and largest diameter in D and store as $d_{\min}$ and $d_{\max}$ respectively.

Compute the out-of-roundness U_r of the cross-section using $d_{\min}$, $d_{\max}$ and d_{nom} .

6.5 Unintended eccentricity tolerance

Fabrication of WTSTs and other metal shells of revolution currently relies upon the circumferential welding of separate cans together. Imperfect cans or misalignment during placement before welding can lead to a misalignment of the shell midsurface known as unintended eccentricity $e_a = e_{\text{tot}} - e_{\text{int}}$ (i.e. difference between the total e_{tot} and intended e_{int} eccentricity). For a WTST where

cans are nominally centre-aligned any eccentricity is unintended ($e_{\text{int}} = 0$). e_a normalised by the average plate thickness at the joint t_{av} yields the unintended eccentricity tolerance as $U_e = e_a / t_{\text{av}}$, the data for which can be obtained from a Stage 6 surface. Each joint in the surface is represented by two adjacent points which have the same vertical coordinate. These points must be corrected using the shell thickness to obtain the midsurface coordinates. The difference between the radial midsurface coordinates yields e_{tot} (which equals e_a when $e_{\text{int}} = 0$). The intended eccentricity e_{int} and average plate thickness at a joint can be determined from the relevant design documents. It is important to note that the unintended eccentricity assessment results are dependent on the quality of surface inpainting in Stage 5. The procedure is given in pseudocode in Algorithm 9, where the returned signed value U_e should be used as $|U_e|$ when comparing against fabrication tolerance limits.

Algorithm 9: Unintended eccentricity tolerance measurement

Input: Measured outer-surface radial coordinates immediately below and above each joint, corresponding nominal outer radii, and plate thicknesses.

Output: Unintended eccentricity tolerance $U_e(\theta)$

for each joint **do**

 Extract the coincident circumferential rows immediately below and above the joint.
 Convert their measured outer-surface radii to midsurface radii by subtracting half the corresponding plate thickness.

 Convert the corresponding nominal outer radii to midsurface radii.

 Compute the total eccentricity: $e_{\text{tot}}(\theta) = \rho_{\text{m,below}}(\theta) - \rho_{\text{m,above}}(\theta)$

 Compute the intended eccentricity: $e_{\text{int}} = r_{\text{m,below}} - r_{\text{m,above}}$

 Compute the unintended eccentricity: $e_a(\theta) = e_{\text{tot}}(\theta) - e_{\text{int}}$

 Compute the average joint thickness: $t_{\text{av}} = (t_{\text{below}} + t_{\text{above}}) / 2$

 Compute the tolerance: $U_e(\theta) = e_a(\theta) / t_{\text{av}}$

end for

return U_e

7. Best practice guidelines on the application of the protocol

The six-stage protocol developed in this paper is purposefully modular to permit methods to be developed for each stage in isolation from all others, it being always clear what the purpose of each stage is and what is important from the point of view of obtaining a ‘shells-quality’ final load-bearing midsurface. We offer the following guidelines on its application:

- Ensure that the ‘Stage 0’ recommendations to the scanning contractor are respected to maximise the quality of the data coming into the protocol.
- The released workflow retains outputs at Stages 1, 2, 5 and 6 to permit fast restart from those cached stages, particularly where the implemented methods still require visual quality control.

- Researchers working on the protocol are encouraged to maintain a consistent interface between stages and separate memory spaces.
- The most frequent starting point for operations will be the *start* of Stage 6 of the protocol where the inpainted representation of the structural midsurface is now available as a finely spaced regular rectangular grid based on the output from Stage 5 (i.e. obtainable via a thickness offset from the outer surface data). It is this grid that is subsequently projected onto a target grid for further analysis, for example a (not necessarily rectangular) grid of nodal positions corresponding to a finite element mesh or a coarser (but still rectangular) grid of coordinates for application of the tolerance checking algorithms of Section 6 or Fourier decomposition methods for further data exploration.
- The ideal output from Stage 6 of the protocol is a data structure that can be directly used to generate an input file with node coordinates, mesh connectivity and sets for load and boundary condition specification.
- Attempting a GMNIA of a gDT obtained at the *end* of Stage 6 must always be accompanied by best practices for advanced finite element analysis of this class of structure, many of which are detailed in (82). For example, model correctness must be verified by a linear elastic analysis (usually denoted as LA, but if deployed on the gDT geometry it would in fact be an LIA) with a manual comparison of base reactions and tip deflections against simple formulae.

8. Protocol limitations

The main limitations of the protocol and its current implementations are set out as follows:

- The current implementations of the six stages of the protocol are intended primarily as a proof of concept of the integrity of this data processing pipeline. Each stage is accompanied by a detailed explanation of the concept behind it as informed by the mechanics and constructability of a large-scale multi-segment metal civil engineering shell structure.
- For the ReleaseTower dataset, excluding the one-time PTS-to-BIN conversion, a complete execution of the released MATLAB implementation took less than one hour on the workstation used in this study; runtime will depend on point count, hardware and parallel configuration. It is *intended* that both the methods set out herein and their implementations be further developed and optimised.
- The Stage 0 recommendations are intended to minimise the risk that the data going into the protocol are deficient to the point that the raw point cloud cannot ultimately be converted into a usable high-quality Stage 6 structured outer-surface representation, from which a structural midsurface may be derived using the plate-thickness data. Currently the most

crucial issue regarding the incoming raw data pertains to having a sufficient overlap of points from adjacent scanning positions (assuming, as will likely be most often the case, that a shell of revolution is scanned from the outside).

- If there is insufficient overlap, registration cannot occur and a full circumference cannot be reconstructed. While this ultimately precludes the development of GMNIA-quality gDT finite element models based on the data, the application of the tolerance-verification algorithms can still be done at a localised level. In such a case, it is necessary to bypass Stages 1 and 5.
- A quantification of error propagation through the stages of the protocol and ultimately in the definition of the final load-bearing shell midsurface remains a formidable challenge. Our preferred approach would be to begin with multiple repeat scans of the same structure, using the repeated measures to establish a ‘mean’ point cloud or midsurface and their associated standard errors, and to attempt to correlate these errors with environmental conditions during scanning. The establishment of an authoritative ‘ground truth’ relative to which absolute errors in geometry can be assessed is unlikely to ever be possible, which will pose a major challenge to any validation endeavour.
- The protocol is designed to create geometrically high-fidelity digital representations of a metal shell of revolution and does not currently accommodate any material considerations such as the self-equilibrating residual stress state that must exist in tandem with the observed geometry. The jury is out on the actual structural significances of this residual stress state, although methods could be developed to approximate the effect of these by enforcing a complete displacement state on a ‘perfect’ geometry to reproduce an observed Stage 6 gDT geometry in tandem with appropriate nonlinear material models.
- The protocol is envisaged for shells of revolution that share a similar rolled isotropic metal welded construction to wind turbine support towers. This should include applications such as monopiles, piles, chimneys, silos and tanks. Externally stiffened or corrugated construction is likely to require additional tailored protocol stages that have not yet been implemented.
- The development of the protocol has been informed by data gathered by terrestrial laser scanning conducted on a structure at the last possible moment onshore before its installation offshore that does not include the temporal dimension necessary for structural health monitoring. A qualitatively identical pipeline should be necessary for data gathered for an offshore structure scanned in-situ should this ever be undertaken, for example with drone-based photogrammetry methods. This would be a major step towards continuous surface measurement and a path towards a spatially and temporally continuous digital twin.

9. Summary

This major paper has focused on the development of a standardised protocol for the creation of geometric digital twins (gDTs) of metal structures that are nominally shells of revolution. The following is set out in detail:

- A detailed literature review tracks the development of reality capture hardware and data processing methods starting in the mid-20th century, where ‘imperfection surveys’ were first performed on thin-walled shell structures using laborious contact methods, up to the present day where automated optical methods permit the gathering of great quantities of high-quality data such that the routine creation of gDTs is now feasible.
- A number of historical trends are identified from the review which are then used to inform the proposed gDT protocol, including the need for high-quality registration of point clouds obtained from multiple scanning positions, the effective obsolescence of Fourier analysis as a core part of the data processing pipeline, and the emergence of gDT creation as a problem requiring HPC-grade computational resources.
- A six-stage automated software protocol serving as a framework for the systematic creation of gDTs of shells of revolution is presented in detail. Artefacts introduced through inadequate data processing can have a disproportionately pathological influence on the quality of the final geometry, and the conception of each stage of the protocol has been informed by the unique sensitivity of a thin-walled shell of revolution to the quality of the geometric definition of the load-bearing midsurface. ‘Best practice’ recommendations to scanning contractors on maximising the quality of data going into the protocol are given.
- The protocol encompasses enhanced point cloud registration (to avoid midsurface discontinuities caused by artificial gaps in overlapping point clouds), can segmentation (to logically partition the structure according to regions of constant plate thickness), systematic and random outlier removal (to further filter any features that may cause artificial midsurface discontinuities), surface reconstruction (to convert an outlier-free registered point cloud into a gridded surface representation), surface inpainting (to permit an exploitation of larger regions of the point clouds and create ‘larger’ gDTs) and projection to arbitrary meshes (for subsequent structural analysis or geometric tolerance assessment).
- The protocol is currently implemented using established candidate algorithms to illustrate effectiveness. These algorithms are illustrated on genuine laser scan data, presented in pseudocode format and released as a functional MATLAB implementation. The individual stages of the protocol and their implementations are actively being developed further as part of an ongoing long-term research programme at Imperial College London.

- Additionally, entirely novel algorithms for the automated and full-structure assessment of buckling-relevant construction tolerances of EN 1993-1-6 (2025) (65) (meridional and circumferential dimple, out-of-roundness and unintended eccentricity) are presented in pseudocode and as MATLAB implementations deployable on a final-stage high-quality surface corresponding to a viable gDT.

MATLAB implementations of the six stages of the protocol and tolerance verification algorithms may be downloaded from:

https://github.com/SadowskiAJ/DTP_V1.git

A 5.7 GB point cloud derived from a laser scan of an actual modern WTST (83) which is used in conjunction with the above MATLAB files, and which users may freely use (with appropriate attribution) to develop their own algorithms and implementations may be downloaded from:

<https://doi.org/10.6084/m9.figshare.28903136.v1>

This dataset is released with the strict caveat that it has been modified to maintain anonymity of the facility and does *not* represent the actual constructed geometry of any one tower structure.

CRedit authorship contribution statement

Lijithan Kathirkamanathan: Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualisation. **Adam Jan Sadowski:** Conceptualisation, Methodology, Validation, Writing - Original Draft, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Marc Seidel:** Conceptualisation, Methodology, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

Acknowledgements

This research was sponsored by the UK Engineering and Physical Sciences Research Council (EPSRC) via an ICASE PhD Studentship award for Lijithan Kathirkamanathan (Voucher No. 2583705, co-sponsored by Siemens Gamesa Renewable Energy) and an Open Plus Fellowship for Adam J. Sadowski (EP/Z533622/1). All contributions are very gratefully acknowledged.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of the final revision, the authors used OpenAI Codex to assist with language refinement, consistency checking between the manuscript, pseudocode and released MATLAB

1331 implementation and code-quality review. All AI-assisted output was critically reviewed and edited
1332 by the authors, who take full responsibility for the content of the manuscript and released software.

1333 References

- 1334 (1) Arbocz J, Abramovich H. *The initial imperfection data bank at the Delft University of*
1335 *Technology: Part I*. Delft University of Technology, Netherlands: Report number: LR-290, 1979.
- 1336 (2) Sadowski AJ, Morata MT, Kathirkamanathan L, Seidel M, Rotter JM. On the existing test
1337 dataset of isotropic cylindrical metal shells under axial compression and the design of modern
1338 metal civil engineering shells. *Structural safety*. 2023; 102 102285.
1339 10.1016/j.strusafe.2022.102285.
- 1340 (3) Koiter WT. *On the stability of elastic equilibrium*. Delft University of Technology; 1945.
- 1341 (4) Singer J, Abramovich H. The development of shell imperfection measurement techniques.
1342 *Thin-walled structures*. 1995; 23 (1-4): 379–398. 10.1016/0263-8231(95)94361-V.
- 1343 (5) Wenk Jr. E, Slankard RC, Nash WA. Experimental analysis of the buckling of cylindrical
1344 shells subjected to external hydrostatic
1345 pressure. *Proceedings of the Society for Experimental Stress Analysis*. 1954; 12 (1): 163–180.
- 1346 (6) Arbocz J. *The effect of general imperfections on the buckling of cylindrical shells*. PhD.
1347 California Institute of Technology; 1968.
- 1348 (7) Arbocz J, Babcock Jr. CD. *Experimental investigation of the effect of general imperfections*
1349 *on the buckling of cylindrical shells*. NASA. Report number: CR-1163, 1968.
- 1350 (8) Arbocz J, Babcock Jr. CD. The effect of general imperfections on the buckling of cylindrical
1351 shells. *Journal of Applied Mechanics*. 1969; 36 (1): 28–38. 10.1115/1.3564582.
- 1352 (9) Arbocz J, Williams JG. Imperfection surveys on a 10-ft-diameter shell structure. *AIAA*
1353 *Journal*. 1977; 15 (7): 949–956. 10.2514/3.7389.
- 1354 (10) Singer J, Abramovich H, Yaffe R. *Initial imperfection measurements of integrally stringer-*
1355 *stiffened cylindrical shells*. Department of Aeronautical Engineering, Technion - Israel Institute
1356 of Technology, Haifa, Israel. Report number: TAE No. 330, 1978.
- 1357 (11) Dancy R, Jacobs D. *The initial imperfection data bank at the Delft University of Technology,*
1358 *Part II*. Delft University of Technology, The Netherlands. Report number: LR-559, 1988.
- 1359 (12) Klonpe AWH, den Reyer PC. *The initial imperfection databank at the Delft University of*
1360 *Technology, Part III*. Delft University of Technology, The Netherlands. Report number: LR-568,
1361 1989.
- 1362 (13) Klonpe AWH, den Reyer PC. *The initial imperfection databank at the Delft University of*
1363 *Technology, Part IV*. Delft University of Technology, The Netherlands. Report number: LR-569,
1364 1989.
- 1365 (14) Klonpe AWH. *The initial imperfection databank at the Delft University of Technology, Part*
1366 *V*. Delft University of Technology, The Netherlands. Report number: LR-734, 1993.

- 1367 (15) Rhijnsburger MPM. *The initial imperfection data bank at the Delft University of*
1368 *Technology, Part VI*. Report number: Memorandum M-867, 1999.
- 1369 (16) Degenhardt R, Kling A, Bethge A, Orf J, Kärger L, Zimmermann R, *et al*. Investigations on
1370 imperfection sensitivity and deduction of improved knock-down factors for unstiffened CFRP
1371 cylindrical shells. *Composite Structures*. 2010; 92 (8): 1939–1946.
1372 10.1016/j.compstruct.2009.12.014.
- 1373 (17) Wang B, Zhu S, Hao P, Bi X, Du K, Chen B, *et al*. Buckling of quasi-perfect cylindrical
1374 shell under axial compression: A combined experimental and numerical investigation.
1375 *International journal of solids and structures*. 2018; 130-131: 232–247.
1376 10.1016/j.ijsolstr.2017.09.029.
- 1377 (18) Li Z, Zhang Q, Shi F, Wang J, Pasternak H. Geometric properties of steel components with
1378 stability and fatigue risks using 3D-laser-scanning. *Buildings*. 2024; 14 (1): 168.
1379 10.3390/buildings14010168.
- 1380 (19) Castro SGP, Zimmermann R, Arbelo MA, Khakimova R, Hilburger MW, Degenhardt R.
1381 Geometric imperfections and lower-bound methods used to calculate knock-down factors for
1382 axially compressed composite cylindrical shells. *Thin-walled structures*. 2014; 74: 118–132.
1383 10.1016/j.tws.2013.08.011.
- 1384 (20) Lin D, Pervizaj A, Madsen SB, Myers AT. Flexural buckling tests on 1:4 scale wind turbine
1385 tower tubes. *Proceedings of the Annual Stability Conference of the Structural Stability Research*
1386 *Council, San Antonio, Texas: 2024*.
- 1387 (21) Peng X, Lin D, Subramanian S, Pervizaj A, Myers AT, Schafer BW. Collapse modeling of
1388 thin-walled cylinder with measured imperfections. *Proceedings of the Annual Stability*
1389 *Conference of the Structural Stability Research Council, San Antonio, Texas: 2024*.
- 1390 (22) Comité européen de normalisation. EN 1993-1-6. *Eurocode 3: Design of steel structures,*
1391 *Part 1-6: Strength and stability of shell structures*. Brussels: CEN; 2007.
- 1392 (23) Sadowski AJ, Orabi MA, Camara A. VSI on GMNIA validation in computational structural
1393 engineering guest editorial. *Data in brief*. 2024; 55 110555. 10.1016/j.dib.2024.110555.
- 1394 (24) Silvestru V. Experimental and simulation data for point-by-point wire arc additively
1395 manufactured carbon steel bars loaded in uniaxial tension. *Data in brief*. 2024; 53: 110093.
1396 10.1016/j.dib.2024.110093.
- 1397 (25) Sebek RWL. *Imperfection surveys and data reduction of ARIANE interstages I/II and II/III.*
1398 *Ir. thesis*. Delft University of Technology; 1981.
- 1399 (26) Clarke M.J., Rotter J.M. *A technique for the measurement of imperfections in prototype silos*
1400 *and tanks*. School of Civil and Mining Engineering, The University of Sydney, NSW, Australia.
1401 Report number: R565, 1988.
- 1402 (27) Ding XL. *Precise engineering surveying: Application to the measurement of large scale*
1403 *steel silos*. PhD. University of Sydney; 1992.
- 1404 (28) Coleman R, Ding XL, Rotter JM. The measurement of imperfections in full-scale steel silos.
1405 *4th International Conference on Bulk Materials, Storage, Handling and Transportation: 7th*

- 1406 *International Symposium on Freight Pipelines*. Barton, ACT: Institution of Engineers, Australia;
1407 1992. p.467–472.
- 1408 (29) Rotter JM, Coleman R, Ding XL, Teng JG. The measurement of imperfections in cylindrical
1409 silos for buckling strength assessment. *4th International Conference on Bulk Materials, Storage,*
1410 *Handling and Transportation: 7th International Symposium on Freight Pipelines*. Barton, ACT:
1411 Institution of Engineers, Australia; 1992. p.473–479.
- 1412 (30) Teng JG, Lin X, Michael Rotter J, Ding XL. Analysis of geometric imperfections in full-
1413 scale welded steel silos. *Engineering structures*. 2005; 27 (6): 938–950.
1414 10.1016/j.engstruct.2005.01.013.
- 1415 (31) Lotmar W. Die Ausmessung der Ölbehälter der "Raffineries du Rhône" bei Collombey (VS).
1416 *Schweizerische Bauzeitung*. 1964; 82 (43): 749–752.
- 1417 (32) Shmutter B, Ethrog U. Calibration of storage tanks. *Photogrammetric Engineering*. 1971;
1418 37: 261–266.
- 1419 (33) Papo H, Shmutter B. Tank calibration by stereophotogrammetry. *Photogrammetria*. 1978;
1420 34 (3): 101–109. 10.1016/0031-8663(78)90003-0.
- 1421 (34) Faig W. Close-range precision photogrammetry for industrial purposes. *Photogrammetria*.
1422 1981; 36 (5): 183–191. 10.1016/0031-8663(81)90014-4.
- 1423 (35) Ethrog U. Measuring the deformations of storage tanks. *Photogrammetria*. 1986; 40 (3):
1424 299–310. 10.1016/0031-8663(86)90025-6.
- 1425 (36) Hornung U, Saal H. Buckling loads of tank shells with imperfections. *International Journal*
1426 *of Non-Linear Mechanics*. 2002; 37 (4-5): 605–621. 10.1016/S0020-7462(01)00087-7.
- 1427 (37) Pukanská K, Gajdošík J, Marcian M, Bartoš K. Determination of deformations of a high-
1428 capacity tank using the technology of terrestrial laser scanning. *American International Journal*
1429 *of Contemporary Research*. 2014; 4 (4): 38–45.
- 1430 (38) Pukanská K, Labant S, Bartoš K, Gašinec J, Zuzik J. Determination of deformation of high-
1431 capacity tank using terrestrial laser scanning . *Acta Montanistica Slovaca*. 2014; 19 (1): 41–46.
- 1432 (39) Truong-Hong L, Lindenbergh R, Fisk P. Storage tank inspection based laser scanning.
1433 *ICSCEA 2019*. Singapore: Springer Singapore; 2020. pp. 987–996.
- 1434 (40) Truong-Hong L, Nguyen N, Lindenbergh RC, Fisk P, Huynh T. Towards a digital twin of a
1435 storage tank using laser scan data. *The International Archives of the Photogrammetry, Remote*
1436 *Sensing and Spatial Information Sciences*. 2021; XLVI-4/W4-2021: 119–124.
1437 <https://doi.org/10.5194/isprs-archives-XLVI-4-W4-2021-119-2021>.
- 1438 (41) Chisholm NWT. Photogrammetry for cooling tower shape surveys. *The Photogrammetric*
1439 *record*. 1977; 9 (50): 173–191. 10.1111/j.1477-9730.1977.tb00080.x.
- 1440 (42) Makuch M. Laser scanning in diagnostics of geometric imperfections of hyperboloid cooling
1441 towers. *Geomatics, Landmanagement and Landscape*. 2024; (4): 105–122.
1442 10.15576/GLL/193928.

- 1443 (43) Sadowski AJ, Van Es SHJ, Reinke T, Michael Rotter J, (Nol) Gresnigt AM, Ummenhofer T.
 1444 Harmonic analysis of measured initial geometric imperfections in large spiral welded carbon steel
 1445 tubes. *Engineering structures*. 2015; 85: 234–248. 10.1016/j.engstruct.2014.12.033.
- 1446 (44) Jay A, Myers AT, Mirzaie F, Mahmoud A, Torabian S, Smith E, et al. Large-scale bending
 1447 tests of slender tapered spirally welded steel tubes. *Journal of Structural Engineering*. 2016; 142
 1448 (12): 04016136. 10.1061/(ASCE)ST.1943-541X.0001605.
- 1449 (45) Mirzaie F, Myers AT, Jay A, Mahmoud A, Torabian S, Smith E, et al. Imperfection
 1450 measurements to predict buckling behavior of slender steel tubes. *Thin-walled structures*. 2018;
 1451 123: 270–281. 10.1016/j.tws.2017.11.016.
- 1452 (46) Mirzaie F, Myers AT, Jay A, Mahmoud A, Smith E, Schafer BW. Analysis of geometric
 1453 imperfections of spirally welded slender steel tubes. *Thin-walled structures*. 2020; 146: 106447.
 1454 10.1016/j.tws.2019.106447.
- 1455 (47) Cartalas C, van der Hoeven HJC, Klompé AWH. *Guide to the data reduction of imperfection*
 1456 *surveys on circular shells*. Delft University of Technology, The Netherlands. Report number:
 1457 Memorandum M-622, 1990.
- 1458 (48) Lu R, Brilakis I. Digital twinning of existing reinforced concrete bridges from labelled point
 1459 clusters. *Automation in construction*. 2019; 105: 102837. 10.1016/j.autcon.2019.102837.
- 1460 (49) Li M, Feng X, Hu Q. 3D laser point cloud-based geometric digital twin for condition
 1461 assessment of large diameter pipelines. *Tunnelling and Underground Space Technology*. 2023;
 1462 142: 105430. 10.1016/j.tust.2023.105430.
- 1463 (50) Mirzaei K, Arashpour M, Asadi E, Masoumi H, Li H. Automatic generation of structural
 1464 geometric digital twins from point clouds. *Scientific reports*. 2022; 12 (1): 22321.
 1465 10.1038/s41598-022-26307-7.
- 1466 (51) Loverdos D, Sarhosis V. Image2DEM: A geometrical digital twin generator for the detailed
 1467 structural analysis of existing masonry infrastructure stock. *SoftwareX*. 2023; 22: 101323.
 1468 10.1016/j.softx.2023.101323.
- 1469 (52) Saloustros S, Settimi A, Ascencio AC, Gamero J, Weinand Y, Beyer K. Geometrical digital
 1470 twins of the as-built microstructure of three-leaf stone masonry walls with laser scanning.
 1471 *Scientific data*. 2023; 10 (1): 533. 10.1038/s41597-023-02417-3.
- 1472 (53) Drobnyi V, Hu Z, Fathy Y, Brilakis I. Construction and maintenance of building geometric
 1473 digital twins: State of the art review. *Sensors*. 2023; 23 (9): 4382. 10.3390/s23094382.
- 1474 (54) Mafipour MS, Vilgertshofer S, Borrmann A. Automated geometric digital twinning of
 1475 bridges from segmented point clouds by parametric prototype models. *Automation in*
 1476 *construction*. 2023; 156: 105101. 10.1016/j.autcon.2023.105101.
- 1477 (55) Tang P, Huber D, Akinci B, Lipman R, Lytle A. Automatic reconstruction of as-built
 1478 building information models from laser-scanned point clouds: A review of related techniques.
 1479 *Automation in Construction*. 2010; 19 (7): 829–843. 10.1016/j.autcon.2010.06.007.

- 1480 (56) Laefer DF, Truong-Hong L. Toward automatic generation of 3D steel structures for building
1481 information modelling. *Automation in Construction*. 2017; 74: 66–77.
1482 10.1016/j.autcon.2016.11.011.
- 1483 (57) Yang L, Cheng JCP, Wang Q. Semi-automated generation of parametric BIM for steel
1484 structures based on terrestrial laser scanning data. *Automation in Construction*. 2020; 112:
1485 103037. 10.1016/j.autcon.2019.103037.
- 1486 (58) Glaessgen EH, Stargel DS. The digital twin paradigm for future NASA and U.S. Air Force
1487 vehicles. *53rd Structures, Structural Dynamics and Materials Conference: Special Session on the*
1488 *Digital Twin*, Honolulu, USA; 2012.
- 1489 (59) Liu M, Fang S, Dong H, Xu C. Review of digital twin about concepts, technologies, and
1490 industrial applications. *Journal of Manufacturing Systems*. 2021; 58: 346–361.
1491 10.1016/j.jmsy.2020.06.017.
- 1492 (60) Bartels J, Xu R, Märtins D, Schuster D, Rolfes R, Marx S. Digital twins in wind energy:
1493 advancements in the life-time comprehensive digitisation. *15th European Conference on Product*
1494 *& Process Modelling (ECPM 2024)*, Dresden, Germany; 2024.
- 1495 (61) Leon-Medina JX, Tibaduiza DA, Parés N, Pozo F. Digital twin technology in wind turbine
1496 components: A review. *Intelligent systems with applications*. 2025; 26: 200535.
1497 10.1016/j.iswa.2025.200535.
- 1498 (62) Dierksen N, Hofmeister B, Hübler C. The Bayesian pattern search, a deterministic
1499 acceleration of Bayesian model updating in structural health monitoring. *Mechanical systems and*
1500 *signal processing*. 2025; 225: 112259. 10.1016/j.ymssp.2024.112259.
- 1501 (63) Zheng T, Heng J, Song J, Zhou D, Luo Y, Dai K. Prognostic digital twinning of lattice-
1502 tubular hybrid wind turbine towers exposed to deterioration. *Structures*. 2025; 79: 109461.
1503 10.1016/j.istruc.2025.109461.
- 1504 (64) Lin X, Teng JG. Iterative Fourier decomposition of imperfection measurements at non-
1505 uniformly distributed sampling points. *Thin-walled structures*. 2003; 41 (10): 901–924.
1506 10.1016/S0263-8231(03)00041-7.
- 1507 (65) Comité européen de normalisation. EN 1993-1-6. *Eurocode 3: Design of steel structures,*
1508 *Part 1-6: Strength and stability of shell structures*. Brussels: CEN; 2025.
- 1509 (66) Kathirkamanathan L, Sadowski AJ, Seidel M, Schafer BW. Recommendations for surface
1510 imperfection surveys for thin-walled metal wind turbine support towers. *Proceedings of the*
1511 *ASME 2022 4th International Offshore Wind Technical Conference. ASME 2022 4th*
1512 *International Offshore Wind Technical Conference. December 7–8, 2022*. Boston, Massachusetts,
1513 USA: ASME; 2023. 10.1115/IOWTC2022-98659.
- 1514 (67) Soudarissanane S, Lindenbergh R, Menenti M, Teunissen P. Scanning geometry: Influencing
1515 factor on the quality of terrestrial laser scanning points. *ISPRS Journal of Photogrammetry and*
1516 *Remote Sensing*. 2011; 66 (4): 389–399. 10.1016/j.isprsjprs.2011.01.005.
- 1517 (68) Lichti DD. Error modelling, calibration and analysis of an AM–CW terrestrial laser scanner
1518 system. *ISPRS journal of photogrammetry and remote sensing*. 2007; 61 (5): 307–324.
1519 10.1016/j.isprsjprs.2006.10.004.

- 1520 (69) De Vries J. *The imperfection data bank and its applications*. PhD. Delft University of
1521 Technology; 2009.
- 1522 (70) Besl PJ, McKay ND. A method for registration of 3-D shapes. *IEEE Transactions on Pattern*
1523 *Analysis and Machine Intelligence*. 1992; 14 (2): 239–256. 10.1109/34.121791.
- 1524 (71) Segal A, Haehnel D, Thrun S. Generalized-ICP. *Robotics: Science and Systems V, June 28 -*
1525 *July 1, 2009*. Cambridge, Massachusetts: MIT Press; 2009. doi.org/10.15607/RSS.2009.V.021.
- 1526 (72) Chen Y, Medioni G. Object modelling by registration of multiple range images. *Image and*
1527 *Vision Computing*. 1992; 10 (3): 145-155. 10.1016/0262-8856(92)90066-C.
- 1528 (73) Bergevin R, Soucy M, Gagnon H, Laurendeau D. Towards a general multi-view registration
1529 technique. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 1996; 18 (5): 540–
1530 547. 10.1109/34.494643.
- 1531 (74) Huang K, Dong Z, Wang J, Fei Y. Weld bead segmentation using RealSense depth camera
1532 based on 3D global features and texture features of subregions. *Signal, Image and Video*
1533 *Processing*. 2023; 17 (5): 2369–2383. 10.1007/s11760-022-02454-2.
- 1534 (75) Ye G, Guo J, Sun Z, Li C, Zhong S. Weld bead recognition using laser vision with model-
1535 based classification. *Robotics and Computer-Integrated Manufacturing*. 2018; 52 9–16.
1536 10.1016/j.rcim.2018.01.006.
- 1537 (76) Ning X, Li F, Tian G, Wang Y. An efficient outlier removal method for scattered point cloud
1538 data. *PloS ONE*. 2018; 13 (8): e0201280. 10.1371/journal.pone.0201280.
- 1539 (77) Huang Z, Wen Y, Wang Z, Ren J, Jia K. Surface reconstruction from point clouds: A survey
1540 and a benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2024; 46
1541 (12): 9727–9748. 10.1109/TPAMI.2024.3429209.
- 1542 (78) Sadowski AJ, Seidel M. 8-MW wind turbine tower computational shell buckling benchmark.
1543 Part 2: Detailed reference solution. *Engineering Failure Analysis*. 2023; 148: 107133.
1544 10.1016/j.engfailanal.2023.107133.
- 1545 (79) Mittal H, Okorn B, Jangid A, Held D. *Self-supervised point cloud completion via inpainting*.
1546 ArXiv [Preprint] 2021. 10.48550/arXiv.2111.10701.
- 1547 (80) Sadowski AJ. On the advantages of hybrid beam-shell structural finite element models for
1548 the efficient analysis of metal wind turbine support towers. *Finite Elements in Analysis and*
1549 *Design*. 2019; 162: 19–33. 10.1016/j.finel.2019.05.002.
- 1550 (81) Kathirkamanathan L, Sadowski A, Seidel M. Imperfections in realistic metal wind turbine
1551 support towers: a one-at-a-time (OAAT) study. *ce/papers*. 2023; 6 (3-4): 1093–1098.
1552 10.1002/cepa.2352.
- 1553 (82) Sadowski AJ, Seidel M, Al-Lawati H, Azizi E, Balscheit H, Böhm M, *et al.* 8-MW wind
1554 turbine tower computational shell buckling benchmark. Part 1: An international ‘round-robin’
1555 exercise. *Engineering failure analysis*. 2023; 148: 107124. 10.1016/j.engfailanal.2023.107124.

- 1556 (83) Kathirkamanathan L, Sadowski AJ, Seidel M. Modified laser scanned point cloud dataset of a
1557 modern wind turbine support tower [dataset]. figshare. 2025. 10.6084/m9.figshare.28903136.v1.
- 1558 (84) Ding V, Torabian S, Adany S, Yun X, Pervizaj A, Myers AT, Schafer BW. Experimental
1559 results of stability of cylindrical shells under combined bending and torsion. *ASCE Journal of*
1560 *Structural Engineering*. 2025; 151(1): 04024188. 10.1061/JSENDH.STENG-13381.
- 1561 (85) Peng X, Lin D, Adany S, Subramanian S, Pervizaj A, Myers AT, Schafer BW. Collapse
1562 simulations of can-welded steel cylinders with measured imperfections under pure bending. *ASCE*
1563 *Journal of Structural Engineering*. 2026; 152(9): 04026115. 10.1061/JSENDH.STENG-15447.
- 1564 (86) Lin D, Peng X, Adany S, Myers AT, Schafer BW. Data associated with the publication:
1565 Collapse simulations of can-welded steel cylinders with measured imperfections under pure
1566 bending [dataset]. Johns Hopkins Research Data Repository. 2026.
1567 <https://doi.org/10.7281/T1PCD9QA>.

1568